

The two instances together make the contrast sharp. At Instance 1, SHAP’s attribution budget $f(x^*) - \mathbb{E}[f(X)]$ is non-zero and SHAP returns non-trivial numbers; at Instance 2 the budget vanishes and every cell collapses. PCI returns the same structural verdicts on all seven desiderata across the two instances. The lesson is not that SHAP gives the wrong answer at the baseline: it correctly reports that the factual prediction does not deviate from its mean. The point is that this is a different question from causal responsibility for y^* : when the prediction equals its baseline, SHAP has no budget to distribute even when the PSCM’s causal mechanism is perfectly intact. PCI asks the responsibility question directly, and answers it whether or not the prediction happens to coincide with its population mean.

4.5 Summary Across Both Examples

The failures above are not artifacts of a particular feature set or parameter choice: each traces to a structural limitation of the SHAP/Causal SHAP framework. Causal SHAP strictly improves on plain SHAP for one desideratum (D-YX), by replacing observational conditioning with interventional distributions; the remaining failures share two sources: prediction models include non-causal features that the conditioning distribution cannot exclude (D-YM, D-MX), and Shapley averaging cannot distinguish direct from indirect causal paths (D-MXY). The OBCB analysis of Section 4.2 exhibits the same family in different surface form: plain SHAP fails **D-A-rank** on the 2-feature game, and Causal SHAP either inherits that failure (Alice) or becomes undefined outright (Bob) once the mediator is admitted as a feature. Across both examples, neither SHAP variant reliably satisfies the desiderata that motivated PCI in Section 3. PCI addresses these gaps through three ingredients the SHAP framework systematically lacks: the witness mechanism, the suspect-set distribution Γ_s , and the use of the realized outcome rather than the prediction baseline; the path-noise asymmetry in the sufficiency world produces a structural gap on D-MXY, which appears already in the witness-free single game and is amplified into a robust margin under a witness rule.

Where we go next. The two examples in this section are small enough to inspect every desideratum individually, but they cover only two causal patterns. Section 5 stress-tests PCI on a synthetic structural model carrying three textbook archetypes — linear necessary-and-sufficient, overdetermined-not-necessary, and preempted (gated) — plus an irrelevance control, and verifies that PCI separates them at a Monte Carlo budget small enough to bootstrap honest error bars. Section 6 then contrasts PCI with gradient-based attribution on a continuous credit-limit example.

5 Synthetic Evaluation with Overdetermination and Undercutting

Where we are. Section 4 has compared PCI with SHAP and Causal SHAP on two small examples and shown where the SHAP framework’s desiderata fail. Those examples cover only two causal patterns: overdetermination through a gating mediator (OBCB) and correlated mediation (signal). This section stress-tests PCI across a wider catalogue of patterns, with an analytic ground truth available throughout: the question is whether PCI correctly separates linear, overdetermined, and preempted contributions on a single structural model, and stays at the noise floor on a variable that plays no role.

To evaluate PCI in a setting that exhibits multiple distinct causal patterns simultaneously, we construct a synthetic structural causal model carrying three archetypes — linear necessary-and-sufficient, overdetermined-not-necessary, and preempted (gated) — together with an irrelevance control.¹⁷

The model has six independent root variables and three deterministic branches feeding the outcome E (the generic outcome Y of Section 3, written E here to match the model diagram; its factual value

¹⁷The classical motivating example is Sally and Billy throwing stones at a bottle: Sally’s stone arrives first and shatters it; Billy’s would have done the same had Sally’s not preempted it. The case is canonical in the actual-causality literature [Halpern, 2016] and motivates the preempted archetype below; we use generic letters rather than personal names to keep variable labelling tight.

is y^* and its necessity/sufficiency worlds Y^n, Y^s):

$$\begin{aligned} L_1, L_2 &\sim \mathcal{N}(0, 1), \quad O_1, O_2 \sim \mathcal{N}(1, 1), \quad P, D \sim \mathcal{N}(0, 1), \\ \text{lin} &= 5L_1 + 10L_2, \\ \text{od} &= \max(5O_1, 5O_2), \\ \text{gate} &= \mathbb{1}\{|L_2| \leq \tau\}, \\ \text{p_branch} &= 5P \cdot \text{gate}, \\ E &= \text{lin} + \text{od} + \text{p_branch}. \end{aligned}$$

We pick $\tau = 0.674$ so the gate is on / off about half the time ($\Pr(|L_2| \leq 0.674) \approx 0.5$ for $L_2 \sim \mathcal{N}(0, 1)$). The DAG is shown in Figure 1. The variable D is sampled but never enters E , serving as an irrelevance control, so any non-zero score for D is the noise floor of the procedure.

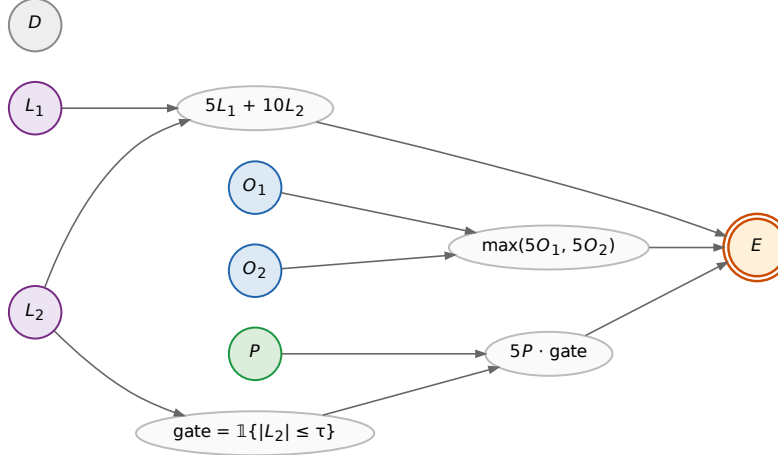


Figure 1: Synthetic archetype model. Roots (circles) are independent draws: $L_1, L_2 \sim \mathcal{N}(0, 1)$ (linear, purple); $O_1, O_2 \sim \mathcal{N}(1, 1)$ (overdetermined via max, blue); $P \sim \mathcal{N}(0, 1)$ (gated by $|L_2| \leq \tau$, green); $D \sim \mathcal{N}(0, 1)$ (irrelevance control, grey). Mediators (white ellipses) carry the deterministic functions of their parents. Outcome E (orange double-circle) is the sum $\text{lin} + \text{od} + \text{p_branch}$. The variable D has no outgoing edges by construction.

The PCI evaluation treats the six roots as the suspect set; deterministic mediators are not suspects. Restricting suspects to roots loses no causal information here since downstream values are determined by the roots (formal statement deferred to Section 3).

Causal impact, decomposed. We follow the impact function $ci(y^s, y^n, y^*) = |y^n - y^*| - |y^s - y^*|$ defined in Section 3, but report its two components separately as ci_N and ci_S :

$$\begin{aligned} ci_N(V) &= \mathbb{E}[|Y^n - y^*| \mid V \text{ intervened}] \quad (\text{necessity}), \\ ci_S(V) &= -\mathbb{E}[|Y^s - y^*| \mid V \text{ pinned at factual}] \quad (\text{sufficiency}), \end{aligned}$$

with $ci(V) = ci_N(V) + ci_S(V)$. Higher ci_N means more necessary; higher ci_S (closer to zero) means more sufficient. The raw scores carry a co-intervention baseline that does not vanish for D , so we report *excess* over D within each case:

$$\Delta_N(V) = ci_N(V) - ci_N(D), \quad \Delta_S(V) = ci_S(V) - ci_S(D).$$

D is the noise floor; any other variable's Δ is its causal signal above it.

Two factual cases. We instantiate the model with one batch of 500 events and select two factual observations to read all archetypes off.

Case 1 (preempted, contestable O pair). Filter: $|L_2| > \tau$ (gate off, so $\text{p_branch} = 0$ regardless of P) and $|O_1 - O_2| < \delta$ with $\delta = 1$ (the two overdetermined contributors are close: the winner is barely the winner; we call such a pair *contestable*, since a small perturbation to either contributor

could flip which one wins $\max(5O_1, 5O_2)$). Linear N+S on the L_s , between-variable sufficiency asymmetry on a contestable O pair, and noise floor for both P and D should all be readable from this single observation.

Case 2 (unpreempted, dominant O winner). Filter: $|L_2| \leq \tau$ (gate on, P branch active) and $|O_1 - O_2| > \delta$. The dominant winner is what makes the within-variable S -not- N gap visible: the winner is large enough that alternatives rarely beat it (high Δ_S), and the loser provides a partial floor when the winner is intervened (reduced Δ_N). A contestable pair cannot show this within-variable gap, since pinning either one still lets the partner flip the max.

The two cases together test the structural-role-not-identity claim: the same variable P should sit at the noise floor in Case 1 and mirror L_1 in Case 2.

Desiderata and measurements. Table 9 pairs each archetype expectation with the formal inequality on the excess scores Δ_N, Δ_S , the measured value at the production sample budget ($N = 20\,000$ Monte Carlo draws per intervention), and a bootstrap 95% confidence interval (4 000 resamples, joint across V and D so per-sample correlations are preserved). The noise-floor variable D itself drifts across the two cases: co-intervention variance is structurally larger in the gate-on regime than in the gate-off regime, so $\text{ci}_N(D)$ shifts between regimes by a non-negligible amount. This is exactly why every archetype claim is stated on the excess scores Δ_N, Δ_S rather than raw ci values: subtracting D within each case normalises out the regime-dependent floor, so the archetype patterns can be read off stably regardless of D 's absolute drift.

The threshold from the data, not from a guess. The directional desiderata of the form $\Delta > \epsilon$ or $|\Delta| < \epsilon$ all depend on a single tolerance ϵ . Rather than hand-pick it, we derive it from the Monte Carlo budget. For every Δ score appearing in any desideratum we bootstrap a standard error σ_V and take

$$\epsilon = 1.645 \cdot \sigma_{\max}, \quad \sigma_{\max} = \max_{V, \text{case}} \sigma_V,$$

the one-sided 95% Z-quantile times the largest standard error across all measured Δ s. Under the null hypothesis $\Delta = 0$, fewer than 5% of measurements would exceed ϵ ; a Δ that clears ϵ is therefore not Monte Carlo chatter at the chosen budget. At $N = 20\,000$ samples we measure $\sigma_{\max} \approx 0.092$, giving $\epsilon \approx 0.151$. Smaller budgets give a noisier ϵ that defeats the point (e.g. at $N = 5\,000$, $\sigma_{\max} \approx 0.18$ would push ϵ into the signal range); this fixes the production sample size for the section. Ten of the twelve desiderata listed in earlier drafts survive this honest threshold; the two that do not (cross-regime P flip, and quantitative match of $\Delta_N(P)$ to $\Delta_N(L_1)$ in Case 2) are discussed below as diagnostics.

#	Formal desideratum	Measured	95% CI	Result
1	$\Delta_N(L_2)$ is the largest excess necessity, Case 1	+2.129	[+1.953, +2.300]	✓
2	$\Delta_N(L_2)$ is the largest excess necessity, Case 2	+0.642	[+0.476, +0.813]	✓
3	$\Delta_S(O_w) - \Delta_S(O_l) > \epsilon$, Case 1 (contestable)	+0.347	[+0.216, +0.473]	✓
4	$\Delta_S(O_w) - \Delta_S(O_l) > \epsilon$, Case 2 (dominant)	+0.529	[+0.387, +0.667]	✓
5	$ \Delta_N(O_w) - \Delta_N(O_l) < \epsilon$, Case 1	-0.093	[-0.266, +0.081]	✓
6	$\Delta_S(O_w) - \Delta_N(O_w) > \epsilon$, Case 2 (within-variable S -not- N signature)	+0.487	[+0.256, +0.712]	✓
7	$\Delta_N(L_2) - \Delta_N(O_w) > \epsilon$, Case 2	+1.082	[+0.915, +1.251]	✓
8	$ \Delta_N(P) < \epsilon$, Case 1 (gate off)	+0.112	[-0.058, +0.280]	✓
9	$\Delta_N(L_1) - \Delta_N(P) > \epsilon$, Case 1	+0.670	[+0.499, +0.843]	✓
10	$\Delta_N(P) > \epsilon$, Case 2 (gate on)	+0.250	[+0.079, +0.422]	✓

Table 9: Desiderata satisfaction on the synthetic archetype model. Threshold $\epsilon = 1.645 \cdot \sigma_{\max} \approx 0.151$ is the smallest Δ gap that, at $N = 20\,000$ samples, exceeds Monte Carlo noise at one-sided 95% confidence. Subscripts w/l denote the winner / loser of the $\max(5O_1, 5O_2)$ branch in the given case. Per-row check identifiers and the underlying numerical computations are listed in the companion notebook `responsibility_archetypes.ipynb`.

Figure 2 visualises the same ten claims as a forest plot: each row carries a plain-language statement of the structural expectation; the dot is the point estimate of the relevant Δ score (or difference of Δ scores); the horizontal bar is its bootstrap 95% CI; and the green band is the row's *pass region*,

i.e. the set of Δ values that would confirm that row's claim given ϵ . The pass region depends on the claim type: $\Delta > \epsilon$ claims pass to the right of ϵ ; $|\Delta| < \epsilon$ closeness claims pass inside the strip $(-\epsilon, +\epsilon)$; rank claims (#1 and #2) pass to the right of 0, since they compare $\Delta_N(L_2)$ against the largest competing suspect. A row is counted confirmed when the point estimate lies in its green band; the CI is shown separately so the reader can see when a measurement is statistically tight and when its CI crosses the pass boundary. All ten dots fall inside their pass bands. The tightest row is #10 ($\Delta_N(P) > \epsilon$ in Case 2, gate on): the point estimate $+0.250$ clears $\epsilon \approx 0.151$ by roughly one σ , with its CI lower edge at $+0.079$ still inside the band on the right side of zero.

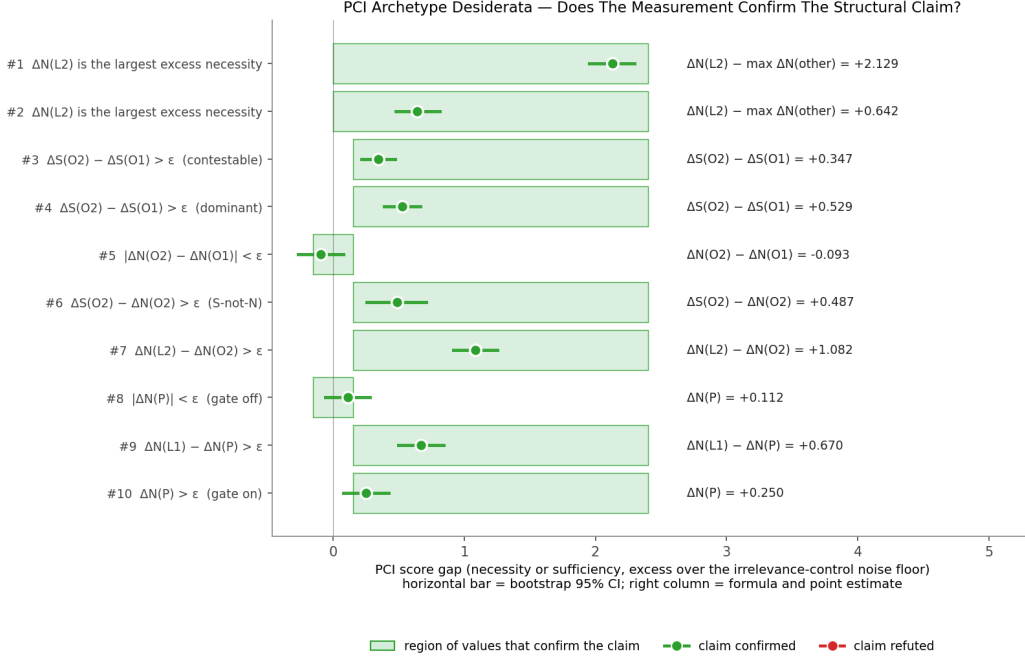


Figure 2: Forest plot of the ten numbered archetype desiderata. Each row carries a plain-language statement of the structural expectation (the case label C1 / C2 marks the factual observation it applies to); the dot is the bootstrap point estimate of the relevant Δ score, the horizontal bar is its 95% CI, and the green band is the row's *pass region*, the set of Δ values that would confirm that row's claim given the bootstrap-derived threshold $\epsilon \approx 0.151$. The right-hand column annotates the formula and its point estimate. All ten dots fall inside their pass bands. Underlying numerical computations are in the companion notebook `responsibility_archetypes.ipynb`.

Diagnostics that do not survive the bootstrap-derived threshold. Two claims from earlier drafts of this section, both about the preempted variable P , fail at the honest threshold and are reported as diagnostics rather than numbered desiderata. (D1) The cross-regime flip $\Delta_N(P, \text{Case 2}) - \Delta_N(P, \text{Case 1})$ measures $+0.139$ with CI $[-0.081, +0.365]$: the CI straddles zero, so P 's rise across regimes is not, on its own, statistically resolved at the production budget. The qualitative content of this claim is already carried by desiderata #8 (P at the noise floor in Case 1) and #10 (P above ϵ in Case 2): if both pass, the regime flip is a corollary, not an independent claim. (D2) The quantitative match $|\Delta_N(P) - \Delta_N(L_1)| < \epsilon$ in Case 2: the signed gap $\Delta_N(P) - \Delta_N(L_1)$ measures -0.267 with CI $[-0.441, -0.106]$, whose magnitude 0.267 is well above ϵ . At this sample budget, gated P does not quantitatively mirror ungated L_1 ; what we can claim is the weaker, qualitative statement that both clear the noise floor in Case 2, which is exactly desideratum #10 for P and built into Case 2 by construction for L_1 .

Where we go next. The archetype evaluation establishes that PCI recovers the right qualitative pattern on every standard causal motif against an analytic ground truth. Section 6 now turns to a complementary continuous-input comparison: how does PCI behave next to gradient-based attribution (Differential Causal Effect of Butler et al., 2022) when the same continuous cause is rescaled or moved out of the steep region of the response surface, where the gradient itself collapses but causal responsibility arguably should not?

6 Comparison to Differential Causal Effect

Having tested PCI against ground truth in Section 5, we now compare it with gradient-based attribution on a continuous-input credit-limit example.

A natural baseline for continuous-input attribution is to read causal influence off the gradient of an interventional mean function. The motivation is direct: $\mathbb{E}[Y \mid \text{do}(X=1)] - \mathbb{E}[Y \mid \text{do}(X=0)]$ is the binary ATE, and for a continuous cause one “arrives at derivative calculus” [Ness, 2025, Section 7.4.3] by letting the contrast shrink: $\partial_x \mathbb{E}[Y \mid \text{do}(X=x)]$. Generalised to a vector $\mathbf{x} = (x_1, \dots, x_d)$ with interventional mean $m(\mathbf{x}) = \mathbb{E}[Y_{\mathbf{X}=\mathbf{x}}]$, the *causal gradient* is

$$\nabla_{\mathbf{x}} m(\mathbf{x}) = \left(\partial_{x_1} \mathbb{E}[Y_{\mathbf{X}=\mathbf{x}}], \dots, \partial_{x_d} \mathbb{E}[Y_{\mathbf{X}=\mathbf{x}}] \right),$$

and each partial is the *differential causal effect* (DCE) of a single variable at $\mathbf{x}$. Butler et al. [2022] introduced this quantity in the Gaussian Process regression setting, where, with the posterior expected mean expressible as $\sum_n k(x, x^{(n)}) \alpha_n$, the DCE has the closed form $\partial_{x_i} \hat{F} = \sum_n \partial_{x_i} k(x, x^{(n)}) \alpha_n$.

PCI and DCE are not direct competitors (DCE estimates a rate of change, PCI estimates individual-level responsibility), but PCI inherits the territory of continuous attribution, so a contrast on a single example is informative. We highlight three differences:

1. DCE requires $y = F(\mathbf{X}, \epsilon)$ with F differentiable; PCI requires only that the causal model be sampleable, and the inference algorithm for the expanded model is the user’s choice (we use Monte Carlo throughout).
2. DCE inherits the units of both $\mathbf{X}$ and Y , so cross-feature comparison is unit-sensitive. The PCI score integrates over the necessity- and sufficiency-world densities, so changes to input scale leave the score essentially unchanged.¹⁸
3. DCE varies abruptly with the local geometry of F . The PCI score depends on where the factual value sits relative to the necessity/sufficiency densities and is therefore non-zero in flat regions while remaining locally informative.

Example 25 (credit-limit assignment). *We model an applicant’s approved credit limit as a function of age and the time of application. The deterministic component is a sigmoidal rise in age with a small sinusoidal modulation in time-of-day; intuitively, age should dominate. We consider four settings on two axes: time measured in hours vs. minutes (a units rescaling that should not affect attribution), and an age prior centred at 30 vs. 40 (a genuine change in what counts as unusual). The deterministic component is shown in Figure 3. Computational details and code are in docs/source/gradient_based_attribution.ipynb.*

PCI scores. We instantiate ci as the absolute-difference score from Example 21, $ci(y^s, y^n, y^*) = |y^n - y^*| - |y^s - y^*|$, and run the search of Section 3 on each setting, conditioning on age and on time of application as singleton suspects. Figure 4 shows the mechanism for the canonical setting (age-30 prior, hours): when age is the active suspect the necessity-world outcome tracks the deterministic curve while the sufficiency-world outcome stays near factual, and the same picture, with a smaller displacement, holds for time of application. Table 10 reports the mean total scores across all four settings. Two patterns stand out. First, age outscores time of application in every setting, by roughly a factor of 3/2, matching the intuition that the age sigmoid moves the outcome by a much larger absolute amount than the daily modulation does. Second, the scores are essentially unchanged between the hours and minutes columns, confirming the unit-invariance discussed above, whereas the shift from the age-30 to the age-40 prior raises every score, since a more unusual factual age means a larger search-space displacement.

PCI vs. DCE across the input grid. Figure 5 contrasts the two methods on the same grid of factual values (age-30 prior, hours); the other three settings give qualitatively the same picture. The

¹⁸This unit-invariance is contingent on how the alternative-value distribution Δ is specified. The ϵ -excised Δ of Section 3 is invariant to a reparameterisation of the input scale precisely when ϵ is fixed by the *probability mass* excised around the factual value (equivalently, in standardised units), as in the ROPE/effect-size reading given there, rather than as an absolute distance in the raw units of $\mathbf{X}$: an absolute-distance ϵ held fixed across a rescaling would itself become unit-dependent and reintroduce the very sensitivity PCI otherwise avoids.

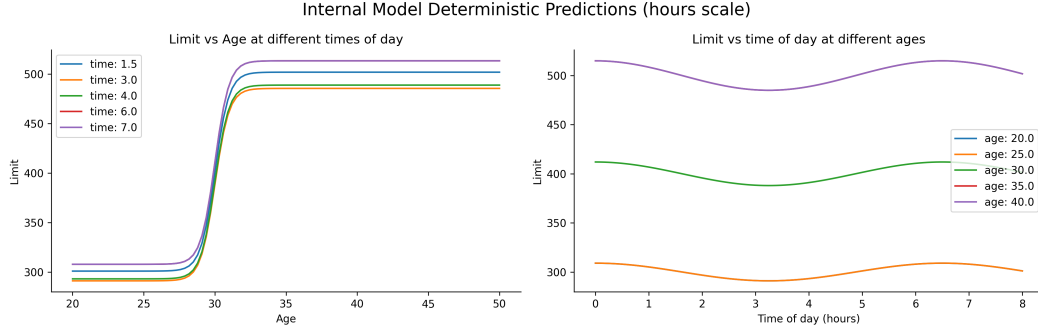


Figure 3: Deterministic component of the credit-limit model (Example 25), on the hours scale: limit vs. age at several times of day (left) and limit vs. time of day at several ages (right). Age dominates the limit, with a small sinusoidal modulation from time of day; the minute scale is identical modulo a linear reparameterisation (verified in the notebook).

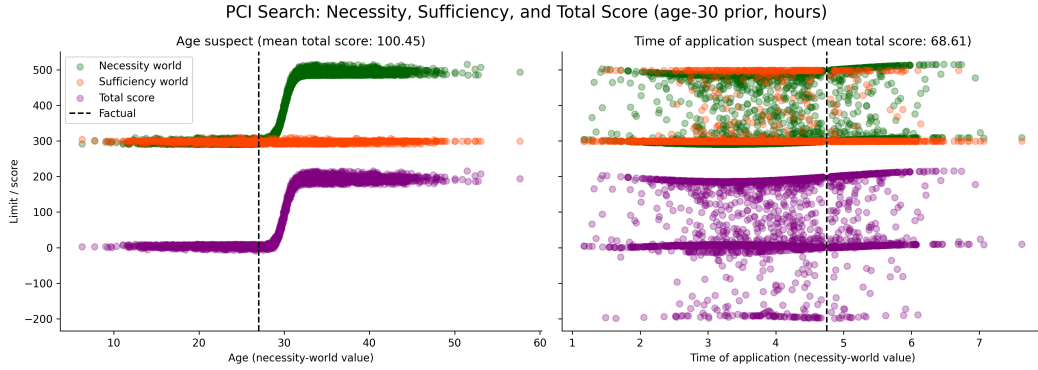


Figure 4: PCI search for the canonical setting (age-30 prior, hours), with age (left) and time of application (right) as the active suspect. Necessity-world (green) and sufficiency-world (orange) limit samples and the total score (purple) are plotted against the necessity-world value of the suspect; the dashed line marks the factual. The necessity world departs from factual along the deterministic response while the sufficiency world stays near it; each panel title gives the mean total score.

top panel plots the difference between the PCI scores for age and for time of application: it is positive everywhere, so age is consistently the more responsible variable, with the gap largest near the steep region of the age-limit sigmoid and growing for ages further from the prior mean. The bottom panels show the DCE for the two inputs, and two differences with PCI stand out. First, on the hour scale the DCE for time of application exceeds the DCE for age over most of the grid and traces a visible sine wave, whereas PCI reports age as the more responsible variable everywhere. Second, switching the time unit from hours to minutes flattens the time DCE by roughly a factor of 60, making the unit sensitivity explicit. Both are direct consequences of DCE’s role as a rate-of-change estimator rather than a responsibility score, and both make it harder to compare across features and across reparameterisations.

Where we go next. Sections 4–6 have positioned PCI against the dominant feature-attribution baselines (SHAP, Causal SHAP, gradient-based DCE) and a synthetic ground-truth catalogue. We now turn to the third leg of the comparison: the formal and empirical link between PCI and *actual causality*. Section 7 states a correspondence theorem identifying, for suitable choices of Γ and Δ , the configurations on which PCI is positive with Halpern’s actual causes; Section 8 compares PCI to Pearl’s *probability of actual causation*, the closest construction in the literature; Section 9 then measures the empirical gap in runtime and verdict recovery between exact AC enumeration and the necessity-only PCI estimator as model size grows.

Table 10: Mean PCI total scores for age and time of application across the four settings (time unit $\times$ age prior), at the factual instance. Age outscores time in every setting; the hours and minutes columns agree (unit-invariance), while the age-40 prior raises all scores relative to age-30 (a more unusual factual age).

Suspect	age-30 prior		age-40 prior	
	hours	minutes	hours	minutes
Age	100.45	97.95	177.87	176.56
Time of application	68.61	66.51	117.97	118.42

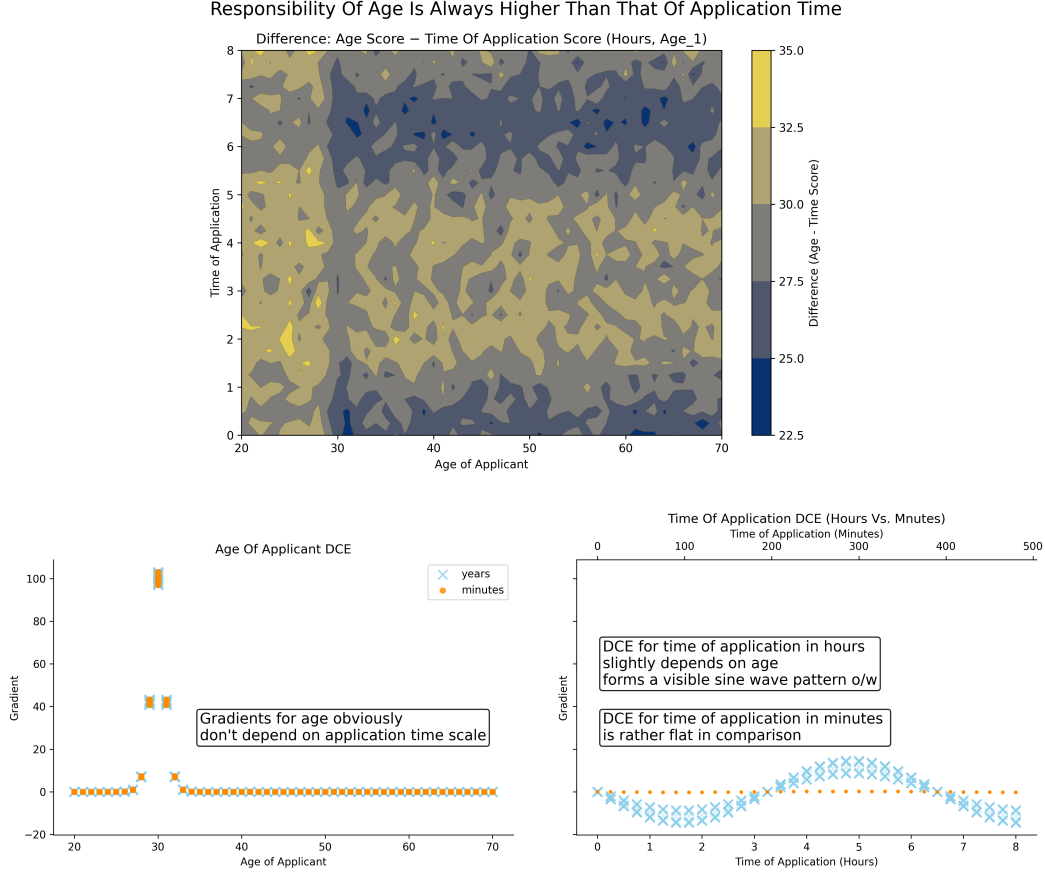


Figure 5: **PCI vs. DCE on the credit-limit grid** (age-30 prior, hours). *Top*: difference between the PCI scores for age and for time of application; positive everywhere, largest near the steep region of the age-limit sigmoid and for ages far from the prior mean, so age is always the more responsible variable. *Bottom*: DCE for age (left) and time of application (right) on both unit scales. Age has the larger absolute DCE only inside the narrow steep region of the sigmoid; elsewhere time of application dominates on the hour scale and goes comparatively flat on the minute scale, illustrating DCE’s shape and unit sensitivity.

7 Relation with Actual Causality

Where we are. Sections 4–6 have positioned PCI against feature-attribution baselines. We now turn to PCI’s relation to the causal-attribution tradition it inherits the witness mechanism from — Halpern’s actual causality. This section establishes the *formal* connection (Theorems 28–29 and Proposition 34); Section 8 adds the side-by-side comparison with Pearl’s *probability of actual causation*; and Section 9 measures the empirical gap in runtime and verdict recovery.

PCI is not meant to explicate Halpern’s notion of actual causality; the goal is to combine the intuitions underlying Halpern’s notion with those motivating Pearl’s probabilities of necessity and sufficiency to provide a more general tool for causal attribution. Nevertheless, to show that we preserve the spirit of Halpern’s definition, we now record the connection between actual causality and the joint necessity-sufficiency machinery of Section 3. The goal is to show that there is a natural choice of variable selection distribution Γ and alternative-value distribution Δ under which PCI verdicts agree with actual-causality verdicts. With one caveat: probability-of-causation maximisation will not identify subset-minimal causes that are not cardinality-minimal, a misalignment we characterise and remedy below.

Scope: a necessity-only correspondence. Halpern’s definition involves factivity, context-sensitive necessity, and minimality, but no sufficiency clause analogous to Pearl’s PS or PNS. The correspondence we establish in this section is therefore *necessity-only*: the sufficiency component of the kernel Φ defined below acts as a factivity check (restoring factual values must preserve φ under fixed noise), not as a substantive sufficiency-attribution component. The substantive sufficiency component of PCI, the Y^s factor in Definition 18 and the ci_S term in Section 5, is exercised in the synthetic overdetermination and undercutting evaluation (Section 5), the differential causal effect comparison (Section 6), and the SIR benchmark (Section 10), all of which are settings where AC cannot adjudicate. The narrowness of this section’s correspondence reflects what AC itself permits, not a limitation of PCI.

We first recall Halpern’s definition.

Definition 26 (Actual Cause, after Halpern, 2016). $C = c$ is an actual cause of φ in the causal setting $(M, \mathbf{u})$ if the following three conditions hold:

- **Factivity:** $(M, \mathbf{u}) \models (C = c)$ and $(M, \mathbf{u}) \models \varphi$,
- **Context-sensitive Necessity:** There is a set of variables $T \subseteq V$ and an alternative setting c' of variables in C such that if $(M, \mathbf{u}) \models T = t^*$, then
$$(M, \mathbf{u}) \models [C \leftarrow c', T \leftarrow t^*] \neg \varphi,$$
- **Minimality:** C is minimal; there is no strict subset $C_{\text{sub}} \subset C$ such that $C_{\text{sub}} = c_{\text{sub}}$ satisfies the above two conditions where c_{sub} is the restriction of c to the variables in C_{sub} .

To connect this to PCI, we instantiate the joint necessity-sufficiency machinery (Definition 18) at a single configuration (C, T) and a fixed noise realisation $\mathbf{u}$, and read off the configuration-level contribution. The resulting kernel combines the same three ingredients as Halpern’s definition: a sufficiency check at factual values, a necessity check against alternatives, and a Γ -weight that records how much the configuration is being asked about.

Definition 27 (AC-aligned PCI kernel). Fix a PSCM $M = \langle U, V, F, P_U \rangle$,¹⁹ a noise realisation $\mathbf{u}$, suspect and witness sets $S, W \subseteq V$ with factual values s^*, w^* , a variable selection distribution Γ on $2^S \times 2^W$ with mass function p^Γ , and an alternative-value distribution $\Delta(s^*)$. For an outcome event φ , active suspect set $C \subseteq S$ and active witness set $T \subseteq W$ with $T \cap C = \emptyset$, define

$$\begin{aligned} \Phi(C, T) &:= p^\Gamma(C, T) \cdot \mathbb{I}[(M, \mathbf{u}) \models [C \leftarrow s^* |_{\mathbf{C}}, T \leftarrow w^* |_{\mathbf{T}}] \varphi] \\ &\cdot \int \mathbb{I}[(M, \mathbf{u}) \models [C \leftarrow c', T \leftarrow w^* |_{\mathbf{T}}] \neg \varphi] \Delta_C(s^*)(dc'). \end{aligned} \quad (18)$$

Concretely, $\Phi(C, T)$ is the matched-configuration slice of the joint necessity-sufficiency measure $P_k^{s,n}$ from Definition 18: integration over P_U is replaced by point evaluation at $\mathbf{u}$, and both the sufficiency and the necessity sums are pinned to the same (C, T) pair, mirroring AC’s structure of testing factivity and the alternative against a single suspect/witness pair. The kernel is positive exactly when

¹⁹The kernel and the theorems below take the full PSCM (structural equations F plus noise distribution P_U) as given, since both the sufficiency and the necessity factors require sampling counterfactual outcomes under $\mathbf{u}$. As discussed in Section 1, this is a stronger commitment than approaches that bound PN/PS/PNS from observational data and a causal graph alone, but is the price of pointwise (not just bound) verdicts at the configuration level.

three conditions hold simultaneously: $(\mathbf{C}, \mathbf{T})$ has positive Γ -mass; restoring $\mathbf{C}$ to its factual values while holding $\mathbf{T}$ fixed preserves φ under $\mathbf{u}$ (the factivity-as-sufficiency component); and there exists a Δ -positive-density alternative $\mathbf{c}'$ for which the corresponding intervention overturns φ under $\mathbf{u}$ (the necessity component). The first factor takes the role played by the suspect/witness biases b_s, b_w in earlier work: regularity of Γ — positive p^Γ on every legal configuration — corresponds to the old $-0.5 < b_s, b_w < 0.5$ regularity. We will say that Γ is *regular* when $p^\Gamma(\mathbf{C}', \mathbf{T}') > 0$ for every $(\mathbf{C}', \mathbf{T}')$ with $\mathbf{C}' \subseteq \mathbf{S}, \mathbf{T}' \subseteq \mathbf{W}, \mathbf{C}' \cap \mathbf{T}' = \emptyset$.

We can now record the forward direction.

Theorem 28. *Suppose that $\mathbf{C}$ is an actual cause of φ with witness set $\mathbf{T}$ in $(M, \mathbf{u})$. Take any $\{\mathbf{S} = \mathbf{s}\} \supseteq \mathbf{C}$ as the potential cause set, subject to the constraint that no element of $\mathbf{S}$ is a deterministic function (under M) of any subset of the remaining elements of $\mathbf{S}$,²⁰ and any $\mathbf{W} \supseteq \mathbf{T}$ as the potential witness set. Assume Γ is regular and that $\Delta(\mathbf{s})$ assigns positive density to every $\mathbf{s}' \in \text{dom}(\mathbf{S}) \setminus \{\mathbf{s}\}$ (positive PMF in the discrete case, positive Lebesgue density in the continuous case). In the continuous case, assume additionally:*

- (i) *the structural equations of M downstream of $\mathbf{C}$ are continuous in $\mathbf{c}'$ under fixed $\mathbf{u}$;*
- (ii) *$\neg\varphi$ is open in $\text{dom}(Y)$ (or, more weakly, the boundary of $\neg\varphi$ has measure zero under the pushforward of Y).*

Then

$$\Phi(\mathbf{C}, \mathbf{T}) > 0, \quad (19)$$

and therefore

$$\sum_{\mathbf{T}' \subseteq \mathbf{W}} \Phi(\mathbf{C}, \mathbf{T}') > 0.$$

Proof. Fix $\mathbf{u}, \mathbf{c}'$, and $\mathbf{T}$ as witnesses to the satisfaction of the conditions in Definition 26. The context-sensitive-necessity clause forces $\mathbf{T} \cap \mathbf{C} = \emptyset$, since otherwise the joint intervention $[\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^*]$ would be ill-defined. It remains to show that each of the three factors in $\Phi(\mathbf{C}, \mathbf{T})$ is strictly positive.

- *Selection weight.* By regularity of Γ , $p^\Gamma(\mathbf{C}, \mathbf{T}) > 0$.
- *Sufficiency indicator.* With exogenous noise $\mathbf{u}$ held constant, factivity gives $(M, \mathbf{u}) \models \varphi$. Restoring $\mathbf{C}$ to $\mathbf{s}^*|_{\mathbf{C}}$ and $\mathbf{T}$ to $\mathbf{w}^*|_{\mathbf{T}}$ leaves the model agreeing with $(M, \mathbf{u})$ on every variable, so the indicator evaluates to 1.
- *Necessity integral.* Context-sensitive necessity provides a specific $\mathbf{c}'$ witnessing $(M, \mathbf{u}) \models [\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^*] \neg\varphi$. The closure condition on $\mathbf{S}$ and the positive-density assumption on $\Delta(\mathbf{s})$ imply that $\mathbf{c}'$ lies in the support of $\Delta_{\mathbf{C}}(\mathbf{s}^*)$. We treat the two cases separately. *Discrete case:* the singleton $\{\mathbf{c}'\}$ has positive PMF $\Delta_{\mathbf{C}}(\{\mathbf{c}'\}) > 0$ by hypothesis, so the integral is at least $\Delta_{\mathbf{C}}(\{\mathbf{c}'\}) > 0$. *Continuous case:* hypothesis (i) gives continuity of the intervention map $\mathbf{c}' \mapsto Y$ at fixed $\mathbf{u}$; hypothesis (ii) places $Y(\mathbf{c}')$ in the interior of $\neg\varphi$;

²⁰Equivalently, $\mathbf{S}$ is closed under no nontrivial structural equation of M . The constraint excludes admitting both S and a deterministic descendant $H = f(S, \dots)$ as candidates in the same suspect set, but covers all candidate sets standardly considered in the AC literature, where structural descendants of one candidate are not admitted as separate candidates: their contribution is mediated by the equation, and AC's minimality clause already prefers the ancestor alone. The same closure condition discharges a joint-support concern: the next assumption requires $\Delta(\mathbf{s})$ to place positive density on every *joint* alternative configuration $\mathbf{s}' \in \text{dom}(\mathbf{S}) \setminus \{\mathbf{s}\}$, not merely positive marginal density on each component. In a chain $S \rightarrow H$ with $H = f(S)$, joint assignments such as $(S=0, H=1)$ are impossible under the structural equations, so any Δ that placed mass on them would be incompatible with M . Closure rules out exactly this case: when no element of $\mathbf{S}$ is a deterministic function of the remaining elements, the joint domain $\text{dom}(\mathbf{S})$ contains no rectangles forced to zero measure by M , and a Δ with positive PMF or Lebesgue density on every $\mathbf{s}' \neq \mathbf{s}$ is consistent with M on the suspect set. In the stone-throwing benchmark of Section 9, the suspect set is $\{A_i, B_i\}_i$ and the deterministic mediators $\{W_i^a, W_i^b\}_i$ live in the witness set, not the suspect set, exactly to keep the suspect set closure-respecting. Witnesses are pinned at their factual values and are not subject to a joint-support requirement on alternatives, so they need no analogous condition.

pulling back an open neighbourhood within $\neg\varphi$ yields an open neighbourhood of $\mathbf{c}'$ on which the indicator equals 1, and this neighbourhood has positive Lebesgue measure and hence positive $\Delta_{\mathbf{C}}$ -mass by the positive-density hypothesis. In either case the integral is strictly positive.

The product of three strictly positive factors is strictly positive, giving $\Phi(\mathbf{C}, \mathbf{T}) > 0$. Summing over $\mathbf{T}' \subseteq \mathbf{W}$ adds non-negative terms to a positive one, preserving positivity. The proof uses factivity and context-sensitive necessity but not minimality: the same conclusion holds whenever $\mathbf{C}$ satisfies factivity and context-sensitive necessity in $(M, \mathbf{u})$, a fact we will use in Proposition 34. $\square$

Remark (when (i)–(ii) bite). In settings with continuous outcomes and an open event such as a threshold $\{Y > y_0\}$ (for instance the SIR overshoot benchmark of Section 10), both hypotheses are satisfied trivially. Binary-output models, where Y takes values in $\{0, 1\}$, satisfy (ii) automatically (every subset of a discrete codomain is open) but may not satisfy (i) when the binary output comes from thresholding a continuous score. In such cases the proof falls back to the discrete branch, which only requires positive PMF on the witnessing $\mathbf{c}'$.

The opposite direction follows by reading the proof in reverse on the necessity factor.

Theorem 29. *Under the same hypotheses on $\{\mathbf{S} = \mathbf{s}\}$, Γ , and Δ as in Theorem 28, if*

$$\sum_{\mathbf{T}' \subseteq \mathbf{W}} \Phi(\mathbf{C}, \mathbf{T}') > 0,$$

then both factivity $((M, \mathbf{u}) \models \varphi)$ and the context-sensitive-necessity condition in Definition 26 hold for $\mathbf{C}$ in $(M, \mathbf{u})$.

Proof. Since Φ is non-negative, the assumption forces $\Phi(\mathbf{C}, \mathbf{T}') > 0$ for at least one $\mathbf{T}' \subseteq \mathbf{W}$. By Definition 27, each factor of $\Phi(\mathbf{C}, \mathbf{T}')$ is strictly positive. The sufficiency indicator equals 1, i.e. $(M, \mathbf{u}) \models [\mathbf{C} \leftarrow \mathbf{s}^* |_{\mathbf{C}}, \mathbf{T}' \leftarrow \mathbf{w}^* |_{\mathbf{T}'}] \varphi$; since pinning variables to their factual values under fixed $\mathbf{u}$ leaves every variable unchanged, this is equivalent to $(M, \mathbf{u}) \models \varphi$, giving factivity. The necessity integral being positive provides a $\Delta_{\mathbf{C}}$ -positive-density set of values $\mathbf{c}'$ for which $(M, \mathbf{u}) \models [\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T}' \leftarrow \mathbf{w}^* |_{\mathbf{T}'}] \neg\varphi$; any such $\mathbf{c}'$, together with $\mathbf{T}'$, witnesses the existential in the context-sensitive-necessity clause of Definition 26. $\square$

Remark (deterministic intermediates and joint support). The hypothesis on $\{\mathbf{S} = \mathbf{s}\}$ is what rules out the following obstruction. Suppose the suspect set were closed under the structural equations of M , e.g., both Sally’s throw S and the bottle’s hit $H = f(S, \dots)$ admitted as candidates. Then the joint assignment $(S=0, H=1)$ is structurally impossible, and any alternative-value distribution consistent with M assigns it density zero, so the joint full-support condition required to recover the witnessing $\mathbf{c}'$ from $\Phi > 0$ fails. Restricting suspects to variables that are not deterministic functions of the others under M removes this obstruction. The constraint is also conservative with respect to the AC literature: in the canonical stone-throwing setup, the candidate set is $\{S_{\text{Sally}}, S_{\text{Bill}}\}$, with H as the outcome rather than a candidate, and AC’s own minimality clause already prefers ancestor candidates over their deterministic descendants when both are admitted. Sections 5 and 9 adopt the same convention experimentally — the continuous synthetic benchmark uses the six roots as suspects and the discrete scaled-throwing benchmark uses $\{A_i, B_i\}_i$ as suspects with the deterministic mediators $\{W_i^a, W_i^b\}_i$ as witnesses — in both cases deterministic intermediates are excluded from the suspect set. Cases where a user genuinely wants to attribute responsibility separately to a deterministic intermediate (rather than to its structural ancestors) lie outside the scope of these theorems and require an alternative reading of intervention semantics; we treat that as a separate problem.

With minimality, the situation is somewhat more complicated. Our choice of Γ will have an impact on which configurations maximise Φ . Moreover, Φ compares configurations by Γ -weight rather than by the subset relation, while Definition 26 formulates minimality strictly in terms of subsets, so some misalignment is to be expected. Some useful observations can be made nonetheless.

Definition 30 (Necessity dilution). *A pair (M, Δ) satisfies necessity dilution on $\mathbf{S}$ if, writing $N(\mathbf{C}, \mathbf{T})$ for the necessity integral in Equation (18),*

$$N(\mathbf{C}^\sharp, \mathbf{T}^\sharp) \geq N(\mathbf{C}^\dagger, \mathbf{T}^\dagger)$$

for every $\mathbf{C}^\sharp \subsetneq \mathbf{C}^\dagger \subseteq \mathbf{S}$ and witnesses $\mathbf{T}^\sharp, \mathbf{T}^\dagger$ disjoint from the corresponding suspect set.

The first is that there is a simplified setting in which a positive value of Φ , under factivity, entails minimality with respect to the same parameters. Three conditions suffice:

- Γ_w is uniform over $2^{\mathbf{W}}$;
- Γ_s is *strictly cardinality-decreasing*, i.e. $p^{\Gamma_s}(\mathbf{C}_1) > p^{\Gamma_s}(\mathbf{C}_2)$ whenever $\mathbf{C}_1 \subsetneq \mathbf{C}_2$ (the modern counterpart of the old null witness bias and positive suspicion bias);
- the pair (M, Δ) satisfies necessity dilution on $\mathbf{S}$ (Definition 30).

The strategy used in the proof entails that we can safely marginalise over witnesses as well.

Theorem 31. *Consider the set of optimisers $\mathcal{O}^* = \arg \max_{\mathbf{C} \subseteq \{\mathbf{S}=\mathbf{s}\}, \mathbf{T} \subseteq \mathbf{W}} \Phi(\mathbf{C}, \mathbf{T})$. Assume $(\mathbf{C}^\dagger, \mathbf{T}^\dagger) \in \mathcal{O}^*$ with $\Phi(\mathbf{C}^\dagger, \mathbf{T}^\dagger) > 0$, that factivity is satisfied for $\mathbf{C}^\dagger$ (as in Definition 26), and that no element of $\{\mathbf{S} = \mathbf{s}\}$ is a deterministic function under M of the others. Let Γ_w be uniform on $2^{\mathbf{W}}$ and let Γ_s be strictly cardinality-decreasing, and assume that (M, Δ) satisfies necessity dilution on $\mathbf{S}$. Then $\mathbf{C}^\dagger$ is an actual cause of φ according to Definition 26.*

Proof. Factivity is assumed; by Theorem 29 the context-sensitive-necessity condition holds for $\mathbf{C}^\dagger$ in $(M, \mathbf{u})$. So the only way $\mathbf{C}^\dagger$ can fail to be an actual cause is by failing minimality. Suppose for contradiction that minimality fails: there is $\mathbf{C}^\sharp \subsetneq \mathbf{C}^\dagger$ satisfying factivity and context-sensitive necessity for φ with some witness $\mathbf{T}^\sharp$. By Theorem 28 (its proof requires only factivity and necessity, not minimality), $\Phi(\mathbf{C}^\sharp, \mathbf{T}^\sharp) > 0$ as well, with the sufficiency indicator and the necessity integral both equal to 1 and a strictly positive value, respectively. Both pairs therefore satisfy $\Phi(\mathbf{C}, \mathbf{T}) = p^\Gamma(\mathbf{C}, \mathbf{T}) \cdot N(\mathbf{C}, \mathbf{T})$.

By the construction of Γ from independent marginals (Section 3),

$$p^\Gamma(\mathbf{C}, \mathbf{T}) \propto p^{\Gamma_s}(\mathbf{C}) p^{\Gamma_w}(\mathbf{T}) \mathbb{I}[\mathbf{T} \cap \mathbf{C} = \emptyset].$$

The $\mathbb{I}[\cdot]$ factor is 1 for both pairs (otherwise the contradiction would have been immediate). Uniformity of Γ_w on $2^{\mathbf{W}}$ makes $p^{\Gamma_w}(\mathbf{T}) = 2^{-|\mathbf{W}|}$ identically, so the witness factor cancels. Strict cardinality-decrease of Γ_s together with $\mathbf{C}^\sharp \subsetneq \mathbf{C}^\dagger$ gives $p^{\Gamma_s}(\mathbf{C}^\sharp) > p^{\Gamma_s}(\mathbf{C}^\dagger)$, and necessity dilution gives $N(\mathbf{C}^\sharp, \mathbf{T}^\sharp) \geq N(\mathbf{C}^\dagger, \mathbf{T}^\dagger)$. Multiplying these (and noting $N(\mathbf{C}^\sharp, \mathbf{T}^\sharp) > 0$ to convert weak into strict in the composite) gives $\Phi(\mathbf{C}^\sharp, \mathbf{T}^\sharp) > \Phi(\mathbf{C}^\dagger, \mathbf{T}^\dagger)$, contradicting $(\mathbf{C}^\dagger, \mathbf{T}^\dagger) \in \mathcal{O}^*$. $\square$

Corollary 32. *Keep the notation and hypotheses of Theorem 31. Consider the set of optimisers $\mathcal{O}^\Sigma = \arg \max_{\mathbf{C} \subseteq \{\mathbf{S}=\mathbf{s}\}} \sum_{\mathbf{T} \subseteq \mathbf{W}} \Phi(\mathbf{C}, \mathbf{T})$. Assume $\mathbf{C}^\dagger \in \mathcal{O}^\Sigma$ with $\sum_{\mathbf{T} \subseteq \mathbf{W}} \Phi(\mathbf{C}^\dagger, \mathbf{T}) > 0$ and that factivity is satisfied for $\mathbf{C}^\dagger$. Then $\mathbf{C}^\dagger$ is an actual cause of φ according to Definition 26.*

Proof. The summation hypothesis entails the existence of some $\mathbf{T}^\dagger \subseteq \mathbf{W}$ with $\Phi(\mathbf{C}^\dagger, \mathbf{T}^\dagger) > 0$. Because the slice $\{(\mathbf{C}^\dagger, \mathbf{T}) : \mathbf{T} \subseteq \mathbf{W}\}$ is contained in the joint search space $2^{\mathbf{S}} \times 2^{\mathbf{W}}$, we have $(\mathbf{C}^\dagger, \mathbf{T}^\dagger) \in \mathcal{O}^*$ as well. The conditions of Theorem 31 are satisfied, so $\mathbf{C}^\dagger$ is an actual cause of φ . $\square$

When necessity dilution holds. The hypothesis is a property of the pair (M, Δ) , not of Γ . It says that adding suspects to a candidate cause does not introduce new Δ -mass of flipping alternatives, and three settings make it automatic. (i) *Single-pivot models*: one variable $X^* \in \mathbf{S}$ is pivotal under $(M, \mathbf{u})$ and the others are structurally idle, so N peaks at $\{X^*\}$ and falls off as more suspects are added. (ii) *Factored Δ without Boolean-OR redundancy*: $\Delta_{\mathbf{C}}$ is a product of marginals $\bigotimes_{X \in \mathbf{C}} \nu_X$ with $\nu_X(\text{dom}(X)) \leq 1$, and the flip event factorises coordinate-wise; $N(\mathbf{C})$ is then a product of factors each at most 1, decreasing in $|\mathbf{C}|$. (iii) *Conjunctive outcome events*: if $\neg\varphi$ requires flipping every variable in a required set $\mathbf{R} \subseteq \mathbf{S}$, the flip region's $\Delta_{\mathbf{C}}$ -mass decreases as $\mathbf{C}$ grows beyond $\mathbf{R}$. The hypothesis fails in disjunctive overdetermination (e.g. $\varphi : A \vee B = 1$ with both A, B factual): no proper subset of $\mathbf{S}$ flips φ , so N is zero on subsets and positive on the full set: the opposite of dilution. In such cases no proper subset of $\mathbf{S}$ is an actual cause anyway, so Theorem 31 is not the appropriate tool; Proposition 34 is.

We can now return to the misalignment we mentioned earlier. One can imagine a reasoning mode in which one looks for causes of an outcome by inspecting the integrated probability $P(\mathbf{C} \rightsquigarrow \varphi) :=$

$\int \sum_{\mathbf{T}} \Phi(\mathbf{C}, \mathbf{T}) P_{\mathbf{U}}(d\mathbf{u})$, not knowing what the values of $\mathbf{u}$ are but knowing that φ occurred. It is worth pointing out that $\arg \max_{\mathbf{C} \subseteq \mathbf{S}} P(\mathbf{C} \rightsquigarrow \varphi)$ might fail to include some $\mathbf{C}$ that is an actual cause of φ for some setting $\mathbf{u}$.

Observation 33. Define $\Xi^{\max} = \arg \max_{\mathbf{C} \subseteq \mathbf{S}} P(\mathbf{C} \rightsquigarrow \varphi)$. It is not the case that if $\mathbf{C}^\dagger$ is an actual cause in $(M, \mathbf{u})$ for some $\mathbf{u}$, $\mathbf{C}^\dagger \in \Xi^{\max}$.

Proof. Consider a model with two binary parents A, B uniformly distributed and a binary child $Y = A \vee B$. Take $\varphi : Y = 1$ and the suspect set $\mathbf{S} = \{A, B\}$ with $\mathbf{W} = \emptyset$. Compare $\mathbf{C}^\dagger = \{A, B\}$ at noise $\mathbf{u}_{11}$ ($A = B = 1$) and $\mathbf{C}^\sharp = \{A\}$ at noise $\mathbf{u}_{10}$ ($A = 1, B = 0$).

- In $\mathbf{u}_{11}$, $\mathbf{C}^\dagger$ is an actual cause: setting both parents to 0 yields $Y = 0$, no proper subset achieves this, and the indicators in Φ are 1. With Γ_w uniform on $2^\emptyset$ (a single configuration of weight 1) and a strictly cardinality-decreasing Γ_s , $\Phi(\mathbf{C}^\dagger, \emptyset) \propto p^{\Gamma_s}(\{A, B\})$.
- In $\mathbf{u}_{10}$, $\mathbf{C}^\sharp$ is an actual cause: intervening to $A = 0$ flips Y to 0, the singleton is minimal, and again the indicators are 1. So $\Phi(\mathbf{C}^\sharp, \emptyset) \propto p^{\Gamma_s}(\{A\})$.

Integrating uniformly over the four noise realisations $\{\mathbf{u}_{ij}\}$, $P(\mathbf{C}^\dagger \rightsquigarrow \varphi) \propto p^{\Gamma_s}(\{A, B\})$ (only $\mathbf{u}_{11}$ contributes) while $P(\mathbf{C}^\sharp \rightsquigarrow \varphi) \propto 2p^{\Gamma_s}(\{A\})$ ($\mathbf{u}_{10}$ and $\mathbf{u}_{11}$ both contribute). Whether $\mathbf{C}^\dagger$ or $\mathbf{C}^\sharp$ ends up in $\Xi^{\max}$ depends on the precise shape of Γ_s : under sharply decreasing Γ_s , $\mathbf{C}^\sharp$ wins; under shallow decrease, $\mathbf{C}^\dagger$ may. Both cannot belong to $\Xi^{\max}$ simultaneously in general, so some actual causes are dropped. $\square$

The reason is structural: actual cause sets may be subset-minimal without being cardinality-minimal, while integrated-probability maximisation is governed by Γ_s , which (under the cardinality-decreasing parameterisation that aligns with AC’s minimality) penalises larger sets. The misalignment is therefore not a defect of Φ but a mismatch between AC’s notion of minimality (subset relation) and Γ -weighted scoring.

The fix is to drop the maximisation step and use Φ directly to enumerate candidate causes, then filter for subset-minimality. The filter recovers exactly the actual causes in $\mathbf{S}$.

Proposition 34. Under the hypotheses of Theorem 28, define

$$\Xi := \left\{ \mathbf{C}^\dagger \subseteq \mathbf{S} : \sum_{\mathbf{T} \subseteq \mathbf{W}} \Phi(\mathbf{C}^\dagger, \mathbf{T}) > 0, \nexists \mathbf{C}' \subsetneq \mathbf{C}^\dagger \text{ with } \sum_{\mathbf{T} \subseteq \mathbf{W}} \Phi(\mathbf{C}', \mathbf{T}) > 0 \right\}.$$

Then Ξ contains exactly the actual causes of φ in $(M, \mathbf{u})$ that are subsets of $\mathbf{S}$.

Proof. Ξ -membership $\Rightarrow$ actual cause. Let $\mathbf{C}^\dagger \in \Xi$. Theorem 29 applied to $\sum_{\mathbf{T}} \Phi(\mathbf{C}^\dagger, \mathbf{T}) > 0$ gives both factivity and context-sensitive necessity. For minimality, suppose $\mathbf{C}_{\text{sub}} \subsetneq \mathbf{C}^\dagger$ satisfied factivity and context-sensitive necessity. By the proof of Theorem 28 (which uses only factivity and necessity, not minimality), we would have $\sum_{\mathbf{T}} \Phi(\mathbf{C}_{\text{sub}}, \mathbf{T}) > 0$, contradicting the no-positive-proper-subset clause of Ξ .

Actual cause $\Rightarrow \Xi$ -membership. Let $\mathbf{C}^\dagger \subseteq \mathbf{S}$ be an actual cause of φ in $(M, \mathbf{u})$. Theorem 28 gives $\sum_{\mathbf{T}} \Phi(\mathbf{C}^\dagger, \mathbf{T}) > 0$. For the no-positive-proper-subset clause: suppose $\mathbf{C}' \subsetneq \mathbf{C}^\dagger$ had $\sum_{\mathbf{T}} \Phi(\mathbf{C}', \mathbf{T}) > 0$. Then by Theorem 29, both factivity and context-sensitive necessity would hold for $\mathbf{C}'$. So $\mathbf{C}'$ would witness a violation of $\mathbf{C}^\dagger$ ’s minimality, contradicting the assumption that $\mathbf{C}^\dagger$ is an actual cause. $\square$

Proposition 34 converts the cardinality–subset misalignment into a procedure: enumerate $\{\mathbf{C} \subseteq \mathbf{S} : \sum_{\mathbf{T}} \Phi(\mathbf{C}, \mathbf{T}) > 0\}$ (which by Theorem 29 are exactly the cause sets satisfying context-sensitive necessity), filter for factivity and subset-minimality, and read off the actual causes. Some idiosyncratic features of the AC notion (in particular, the subset-minimality clause, which interacts poorly with Γ -weighted scoring) are part of what motivated the more scalable PCI formulation of Section 3 in the first place; the filter procedure here is the right tool when AC verdicts specifically are wanted, but expectation-based PCI attributions are what scale to ML-grade models.

Summary of guarantees. The section delivers, under explicit hypotheses on $\mathbf{S}$, Γ , and Δ : (i) a forward direction (Theorem 28) “actual cause $\Rightarrow \Phi > 0$ ”; (ii) a backward direction (Theorem 29) “ $\sum_{\mathbf{T}} \Phi > 0 \Rightarrow$ context-sensitive necessity”; (iii) minimality preservation under cardinality-decreasing Γ_s (Theorem 31, Corollary 32); (iv) a counterexample (Observation 33) showing that integration over noise misaligns with subset-minimality; and (v) a corrective filter procedure (Proposition 34) that recovers AC verdicts exactly. The empirical counterpart of the correspondence, a benchmark of the necessity-only PCI estimator against exact subset enumeration on a scaled stone-throwing problem, is reported in Section 9.

Where we go next. Before turning to the empirical benchmark, the next section locates PCI against the construction in the literature closest to its expectation form: Pearl’s *probability of actual causation*. The two share the same surface idea, “how often, under the noise, does the actual-cause verdict fire?”, and they coincide on standard examples, but they disagree on graded responsibility patterns that PCI registers and Pearl’s binary AC verdict discards. Section 8 walks through Pearl’s canonical desert-traveller illustration and a weak-poison variant to make the distinction concrete.

8 Relation with the Probability of Actual Causality

Now we are in the position to discuss the relation between PCI and Pearl’s probability of actual causation. On Pearl’s Definition 10.3.5 [Pearl, 2009, Ch. 10],

$$P(\text{caused}(x, y \mid e)) = \frac{P(U_{xy} \cap U_e)}{P(U_e)}, \quad U_{xy} = \{\mathbf{u}: \text{AC}(x, y; M, \mathbf{u})\}, \quad (20)$$

is the posterior probability, given the evidence, that we are in a noise state for which the actual-cause verdict holds, with $\text{AC}(x, y; M, \mathbf{u})$ a chosen actual-causation predicate: the causal-beam definition of [Pearl, 2009, Ch. 10] or the Halpern–Pearl AC1–AC3 [Halpern, 2016].²¹ The same construction underwrites Chockler and Halpern’s degree of blame [Chockler and Halpern, 2004], which substitutes a graded responsibility share inside the expectation. Because (20) is the closest construction in the literature to PCI’s expectation in Definition 18, we now ask in detail how it differs from PCI and how PCI improves on it.

We proceed in three steps. We replay Pearl’s canonical desert-traveller illustration and Pearl’s binary actual-cause verdicts; compute PCI on the same example and show it recovers Pearl’s ranking with graded magnitudes; and introduce a weak-poison variant that exhibits three intuitive responsibility patterns, two of which Pearl’s binary verdict cannot capture but PCI’s graded reading does, and one (within-scenario ranking) Pearl preserves. Then we move to discuss other differences between PCI and Pearl’s probability of causation, which make the former more general and more scalable.

8.1 Pearl’s Worked Example: The Desert Traveller

We start with the canonical illustration used by Pearl [Pearl, 2009, §10.3.3].

Example 35 (Desert traveller). *A traveller has two enemies. The shooter (enemy 2) shoots and empties the canteen ($X=1$); the poisoner (enemy 1), unaware, puts cyanide in the water ($P=1$). The traveller dies ($Y=1$). Who is the actual cause of death?*

The example involves one source of uncertainty packed into a binary noise u :

- $u = 0$: traveller drank the poisoned water before the canteen was emptied; cyanide killed him.
- $u = 1$: canteen was empty before he could drink; dehydration killed him.

There are at least two intuitions here that we would like to recover. One, in any particular case, we want to be able to identify the actual cause. The other, perhaps a bit weaker, we still don’t want to

²¹All computations behind the comparison in this section, Pearl’s original desert-traveller and the weak-poison variant, are collected in the companion notebook `docs/source/desert_traveler.ipynb`, which runs the PCI Monte Carlo search of Section 3 on a scenario-specific structural model for each forensic posterior alongside a parallel hand-rolled enumeration; the two agree within Monte Carlo noise on every value reported below.

say that the dormant cause is completely blameless. Actual causality gives us the first half of the story, assigning full responsibility to one source here. Pearl’s beam analysis [Pearl, 2009, Ch. 10] (equivalently Halpern–Pearl AC1–AC3 of Halpern, 2016) gives the actual-cause verdicts at each deterministic state:

- In $u = 0$: $P=1$ is the actual cause; $X=1$ is not.
- In $u = 1$: $X=1$ is the actual cause; $P=1$ is not.

So $U_{x,y} = \{u = 1\}$ and $U_{p,y} = \{u = 0\}$. Two natural forensic reports each push Pearl’s posterior to certainty (Table 11): toxicology finding cyanide in the body is incompatible with $u = 1$, so $P(\text{caused poisoner} \mid e_A) = 1$; a forensic report of no cyanide is incompatible with $u = 0$, so $P(\text{caused shooter} \mid e_B) = 1$.

Forensic posterior	$P(\text{caused shooter})$	$P(\text{caused poisoner})$
e_A : cyanide detected ($u = 0$)	0	1
e_B : no cyanide ($u = 1$)	1	0

Table 11: Pearl’s probability of actual causality $P(\text{caused} \mid e)$ on the basic desert traveller. The verdict saturates at 1 for the cause whose path operated in the forensically identified scenario, and at 0 for the dormant one.

8.2 PCI on the Basic Desert Traveller

To compare directly with Pearl, we run PCI under the same forensic conditioning: one structural model per scenario (endogenous variables $X, P, c, d, Y \in \{0, 1\}$, with c the cyanide path operating and d the dehydration path), with u pinned at the scenario’s forensic realisation. Suspects are $\mathbf{S} = \{X, P\}$ with singleton candidate causes; the witness pool is the mediators $\{c, d\}$ under a cardinality-uniform Γ ; Δ deterministically flips $1 \rightarrow 0$; the non-cause suspect is left to the SCM rather than intervened on; and the noise distribution P_U is the forensic posterior, a point mass at the scenario’s u (we do *not* condition on the outcome Y itself). The full structural model and component specification are given in Appendix C.

For each candidate cause C we report three quantities, all in $[0, 1]$,²² higher meaning more responsibility, jointly averaged over the four sources of randomness (Γ , Δ , P_U , and the SCM-driven non-cause suspect):

$$\begin{aligned}
N_C &= \mathbb{E}[\mathbb{1}\{Y^n \neq y^*\}] && \text{(necessity — removal flips } Y), \\
S_C &= \mathbb{E}[\mathbb{1}\{Y^s = y^*\}] && \text{(sufficiency — holding at factual sustains } Y), \\
J_C &= \mathbb{E}[\mathbb{1}\{Y^n \neq y^*\} \mathbb{1}\{Y^s = y^*\}] && \text{(joint PNS-style).}
\end{aligned} \tag{21}$$

Under the spec above (computation in the companion notebook), the results are:

Forensic posterior	Cause	N	S	J	Pearl $P(\text{caused})$
e_A : $u = 0$ (cyanide)	shooter	1/4	11/12	5/24	0
	poisoner	1/3	1	1/3	1
e_B : $u = 1$ (dehydration)	shooter	1/3	1	1/3	1
	poisoner	1/4	11/12	5/24	0

Table 12: PCI and Pearl on the basic desert traveller, under the scenario’s forensic posterior. PCI’s joint score J ranks the same cause Pearl’s $P(\text{caused})$ does in each scenario; PCI returns a graded reading where Pearl returns the binary $\{0, 1\}$.

²²These are evaluated for a fixed candidate cause C , hence normalised to $[0, 1]$; this differs from the subprobability $\mathbb{E}[c_i]$ of Definition 19, whose total mass is $\Pr_\Gamma[X_k \in \mathbf{C}] \leq 1$ because it sums only over suspect sets that contain X_k .

Three points to read off Table 12:

(1) PCI and Pearl agree on the ranking. In each scenario, J is strictly larger for the cause Pearl identifies, and strictly smaller for the other. PCI’s ordering matches the actual-cause verdict at each forensic posterior.

(2) PCI is graded; Pearl is binary. Pearl’s $P(\text{caused})$ saturates at 0 or 1. PCI’s J never reaches 1 here because the Γ -averaging over witness subsets dilutes each cause’s score: the “best” witness subset for the operative cause does flip the outcome deterministically and gives $J = 1$ at that single configuration, but the cardinality-uniform average over the witness subsets (weights $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$ over $\emptyset, \{c\}, \{d\}, \{c, d\}$) dilutes this to $1/3$; the per-subset breakdown is in the companion notebook.

(3) The dormant cause is not at zero. PCI assigns $5/24 \approx 0.21$ to the dormant cause. This is not noise: in some witness configurations the dormant cause’s removal does flip Y , because the non-cause suspect is marginalised and the model is not artificially pinned to “this scenario” beyond the noise realisation itself. Pearl’s binary indicator throws this graded information away.

8.3 PCI Expectation vs the AC Machinery

The agreement above is reassuring, but the more interesting observation is what changes underneath. AC’s necessity clause is an *existential*: *there exists* a witness set $\mathbf{T}$ and an alternative value c' such that the intervention flips Y . To verify the existential one in principle enumerates the candidate $(\mathbf{T}, c')$ pairs: this is the main source of AC’s combinatorial hardness. Pearl’s definition then averages the indicator of that already-enumerated predicate over noise states, inheriting the enumeration unchanged.

PCI collapses both moves into one expectation: Φ is averaged over the joint $(\mathbf{C}, \mathbf{T}, \mathbf{u})$ under $\Gamma \otimes P_{\mathbf{U}}$. The existential “does *some* witness work?” becomes the quantitative “*what fraction* of witnesses work?”, the information AC uses only one piece of, exposed as a graded score with sampling-based estimation.

In the desert traveller this is small change: four candidate witness sets, all trivially enumerable. The point is the scaling. In a model with n candidate witness variables, AC’s existential demands checking up to 2^n subsets per noise state, and Pearl’s posterior over the AC indicator inherits that cost per evaluation. PCI’s expectation can be estimated at a Monte Carlo budget the user controls, with variance independent of the problem’s combinatorial size. The desert-traveller agreement is the small- n end of a curve whose large- n end is the computational gap that motivates PCI in the first place.

So far we have two conceptual differences: PCI’s graded reading and its Monte Carlo scaling. We have already seen that this helps to recover the intuition that the shooter of a poisoned victim, while less blameworthy is nevertheless not completely blameless. Now we turn to another example, riding on the fact that actual causality is devoid of a sufficiency component, where a small change to the model makes Pearl’s binary indicator structurally unable to track the responsibility patterns we think a responsibility attribution method should capture.

8.4 Weak-Poison Variant: Two Intuitions Pearl Misses

We modify the desert traveller so the cyanide path is unreliable while the dehydration path stays deterministic. The change is small, but it surfaces three intuitions that a graded responsibility metric should capture and a binary actual-cause indicator structurally cannot.

Example 36 (Desert traveller, weak poison). *Same enemies as in Example 35. The cyanide is a small dose that turns out fatal only with probability $\alpha = 0.1$, governed by an independent exogenous coin $\xi \sim \text{Bern}(\alpha)$. The structural equations are*

$$c = p(u' \vee x'), \quad v_C = c \cdot \xi, \quad d = x(u \vee p'), \quad y = v_C \vee d,$$

where primes here denote Boolean complement ($x' = 1 - x$, and likewise u', p'), not the alternative-value c' of the necessity intervention, and v_C is the new “cyanide-was-fatal” mediator: drinking the poison ($c = 1$) only kills when the fatality coin also operates ($\xi = 1$). $\alpha = 1$ recovers Pearl’s original.

Three responsibility intuitions for an attribution method. Before any computation, the three intuitions that we expect an attribution method to recover in this case are:

1. **Adding noise to a path should lower its own cause’s responsibility.** The cyanide path is now unreliable. In the scenario where cyanide killed (Scenario A, $u = 0, \xi = 1$), the poisoner is still the actual cause, but *less* responsible than in the basic deterministic version, because the killing depended on a coin landing favourably.
2. **Within a scenario, the operative cause should still win.** In Scenario A, even after softening, the poisoner should rank above the shooter. In Scenario B, the shooter should rank above the poisoner.
3. **A reliable cause in its own scenario should beat an unreliable cause in its own scenario.** Comparing across scenarios: the shooter in Scenario B (dehydration deterministic) should outrank the poisoner in Scenario A (cyanide stochastic). Their respective paths are operative in their own scenarios, but one is reliable and the other is not.

Pearl’s $P(\text{caused})$ captures (2) (the within-scenario ranking), but misses (1) and (3). The AC predicate fires deterministically once we condition on the forensic posterior: in Scenario A, $u = 0$ and $\xi = 1$ jointly imply the cyanide path was the operative chain, so $P(\text{caused poisoner} \mid e_A) = 1$; in Scenario B, $u = 1$ implies the dehydration chain operated, so $P(\text{caused shooter} \mid e_B) = 1$. Both scenarios saturate at 1 for their respective operative cause, identical to the basic version (Table 13). Pearl’s binary indicator is structurally indifferent to whether the cyanide path is deterministic or stochastic, and indifferent to a cross-scenario comparison.

Forensic posterior	$P(\text{caused shooter})$	$P(\text{caused poisoner})$
e_A : cyanide killed ($u = 0, \xi = 1$)	0	1
e_B : dehydration killed ($u = 1$)	1	0

Table 13: Pearl’s $P(\text{caused})$ on the weak-poison variant. Identical to the basic version (Table 11); the binary AC indicator is insensitive to α .

What PCI does on the same model. We run PCI exactly as in Section 8.2, with one mechanical change: the witness pool becomes $\mathbf{W} = \{c, d, v_C\}$ (the new mediator v_C is included), and the forensic posterior in Scenario A pins both $u = 0$ and $\xi = 1$ (cyanide-was-fatal is directly forensically observable). Scenario B’s posterior pins $u = 1$; ξ is structurally irrelevant since the cyanide path is dormant.

Table 14 reports PCI’s three quantities alongside Pearl’s verdict. The values are computed in the companion notebook and verified by a hand-rolled enumeration matching the framework.

Forensic posterior	Cause	N	S	J	Pearl $P(\text{caused})$
e_A : $u = 0, \xi = 1$ (cyanide killed)	shooter	≈ 0.167	≈ 0.958	≈ 0.146	0
	poisoner	≈ 0.208	1	$\approx \mathbf{0.208}$	1
e_B : $u = 1$ (dehydration killed)	shooter	1/2	1	1/2	1
	poisoner	1/4	≈ 0.750	≈ 0.125	0

Table 14: PCI and Pearl on the weak-poison desert traveller, under the scenario’s forensic posterior. PCI’s J ranks the same cause Pearl’s binary verdict does in each scenario; the magnitudes also reflect the asymmetry between the deterministic dehydration path and the stochastic cyanide path.

The three intuitions are all captured by J . **(1) Adding noise lowers the noisy cause’s responsibility.** The poisoner in Scenario A drops from $J = 1/3$ in the basic version (Table 12) to $J \approx 0.208$ in the weak version (Table 14). The drop comes through the sufficiency factor: when we hold P at its factual value, the witness mechanism averages over configurations where v_C is not pinned and the cyanide step’s ξ coin still acts. Even at $\xi = 1$ in the forensic posterior, the witness integration

sees the fragility, and J falls. **(2) Within a scenario, the operative cause still wins.** In Scenario A, $J_{\text{poisoner}} \approx 0.208 > J_{\text{shooter}} \approx 0.146$. Cyanide was the operative path, and PCI tracks that. In Scenario B, $J_{\text{shooter}} = 1/2 > J_{\text{poisoner}} \approx 0.125$. **(3) Reliable beats unreliable, across scenarios.** The shooter in Scenario B ($J = 1/2$) outranks the poisoner in Scenario A ($J \approx 0.208$). The dehydration path is deterministic; conditional on $u = 1$, holding X at factual sustains Y with probability 1 regardless of the witness regime, so the shooter is a textbook responsible cause. The cyanide path is fragile; conditional on $u = 0, \xi = 1$, the sufficiency factor still sees the ξ coin under unpinned witness regimes, so the poisoner cannot hit the same ceiling. Pearl’s PAC misses the desiderata for the following reasons. (1) requires sensitivity to mechanism noise, which the AC indicator does not have once the forensic posterior is applied; (3) requires cross-scenario comparison of *strength of responsibility*, which Pearl’s $P(\text{caused})$ does not produce because each posterior independently saturates at 0 or 1.

8.5 Further Differences Between Pearl’s Probability of Causation and PCI

Breadth of aggregation. Def. 10.3.5 aggregates over one dimension: the noise $\mathbf{u}$, conditional on evidence e , for a claim $(X = x, Y = y)$ already fixed. PCI aggregates over four: the suspect set $\mathbf{C}$ and witness set $\mathbf{T}$ via Γ , the alternative value $\mathbf{c}'$ via Δ , and the noise via $P_{\mathbf{U}}$. This is what supports per-feature comparison: the same PCI expectation, evaluated over varying $\mathbf{C}$ inside a fixed $\mathbf{S}$, returns the relative responsibility of each candidate cause. Def. 10.3.5 cannot perform that comparison without a separate evaluation per claim, each inheriting the AC-predicate cost discussed above.

Continuous variables. The AC predicates inside Def. 10.3.5 are event-shaped: they read off $\text{AC}(x, y; M, \mathbf{u})$ for propositional x, y , and extensions to continuous variables require discretising the relevant event before the predicate applies. PCI’s *ci* function is continuous-native (the absolute-difference impact score of Section 3), and the synthetic benchmark of Section 5 exercises that continuity directly.

Lineage in Pearl’s PN/PS. Def. 10.3.5 takes a binary actual-causation predicate — itself a structural-equation construction that does not appeal to PN/PS at all — and lifts it to a probability through $P(\mathbf{u} \mid e)$; PN, PS, and PNS appear nowhere in the construction. PCI, by contrast, generalises PN and PS directly: the necessity factor Y^n is the counterfactual operation underlying PN, the sufficiency factor Y^s is the counterfactual operation underlying PS, both indexed by the cause–witness–alternative–noise tuple, with Γ and Δ exposing the design choices PN/PS makes implicitly. So Pearl’s Def. 10.3.5 and PCI are not competing implementations of the same idea: they inherit from different parts of Pearl’s own toolkit, with PCI playing the role of a context-sensitive, multi-variable extension of the PN/PS machinery rather than a probabilistic wrapper around AC.

Pearl’s Def. 10.3.5 and PCI share the same explanatory target, probabilistic responsibility built on top of structural information, but instantiate it through different mechanisms. Pearl’s definition averages a binary AC indicator against a posterior over noise; PCI integrates a continuous necessity–sufficiency kernel against a joint distribution over suspect sets, witness sets, alternative values, and noise. The basic desert traveller shows the two agree on ranking with PCI providing graded magnitudes; the weak-poison variant shows three intuitions that PCI’s graded reading captures and Pearl’s binary indicator structurally cannot. The cases where sufficiency does even more graded work — the continuous signal-mediation walkthrough of Section 4.3, the synthetic archetypes of Section 5, and the SIR benchmark of Section 10 — are where these constructions answer recognisably different questions, and the question PCI answers is the one we need for ML-grade causal attribution.

Where we go next. Sections 7–8 have laid out the formal relation between PCI and actual causality (and its Pearlian probabilistic wrapper). Section 9 now turns to the *empirical* side of that comparison: a generalised stone-throwing problem where exact subset enumeration of actual causes scales as $(2^{2n} - 1)^2$ cause–witness pairs and is therefore the natural pressure test for the necessity-only PCI estimator. We report runtime, search-space size, and attribution-accuracy curves as the problem size grows.

9 Scaling Actual-Cause Computation: An Empirical Comparison

The correspondence in Theorems 28–29 and Proposition 34 is a statement about verdicts: under the stated hypotheses, AC actual causes coincide with the support of $\sum_{\mathbf{T}} \Phi$. This section reports an empirical companion to the correspondence. We ask whether the necessity-only PCI kernel of Definition 27, instantiated with a finite Monte Carlo estimator, recovers AC verdicts at problem sizes where exact subset enumeration is no longer tractable. The full implementation, raw outputs and plotting code are in the companion notebook `docs/source/actual_causality_benchmark.ipynb`.

Generalised throwing problem. For a size parameter n we replicate the canonical Sally/Billy structure n times in parallel. Binary suspects $A_1, \dots, A_n, B_1, \dots, B_n \in \{0, 1\}$ feed deterministic mediators $W_i^a = A_i$, $W_i^b = B_i \wedge \neg A_i$ that capture overdetermination via preemption: B_i contributes only if A_i does not. The outcome is the conjunction $Y = \bigwedge_{i=1}^n (W_i^a \vee W_i^b)$, and the model size grows as $4n+1$ variables. Ground-truth responsibility at site i is read off during the forward pass: A_i is responsible if $W_i^a = 1$, B_i is responsible if $W_i^b = 1$. The structure scales the canonical example without softening it — overdetermination, undercutting, and witness-mediated symmetry-breaking are present at every site — while keeping a closed-form ground truth available for evaluation.

The variable partition takes the roots $\{A_i, B_i\}_i$ as suspects and the deterministic mediators $\{W_i^a, W_i^b\}_i$ as witnesses, with no suspect a deterministic function of the others. This partition satisfies the closure condition required by Theorem 28 (see the Remark on deterministic intermediates and joint support, which addresses precisely the scenario in which both a root and its structural descendant are admitted as suspects). The benchmark is therefore within the scope of the formal correspondence; in particular, joint full support of the alternative-value distribution holds on the admissible suspect set $\{A_i, B_i\}_i$ even though the model itself is deterministic, because witnesses are not themselves part of the alternative-value distribution.

Exact subset enumeration (AC reference). The exact procedure mechanises Definition 26: for each non-empty cause subset $\mathbf{C} \subseteq \{A_i, B_i\}_i$ and each non-empty witness subset $\mathbf{T} \subseteq \{W_i^a, W_i^b\}_i$, we intervene $[\mathbf{C} \leftarrow 1 - \mathbf{c}^*, \mathbf{T} \leftarrow \mathbf{w}^*]$, record whether the outcome flips, and accept $\mathbf{C}$ as actual cause if some witness $\mathbf{T}$ achieves a flip and no proper subset of $\mathbf{C}$ does. The search space has $(2^{2n} - 1)^2$ cause-witness pairs (9, 225, 3969, 65025 for $n = 1, 2, 3, 4$), which already determines the ceiling reached by the exact method.

Approximate PCI estimator. The PCI estimator instantiates the kernel of Definition 27 via the Monte Carlo search procedure of Section 3. Algorithm 1 is the explicit procedure used to produce the curves in Figures 6–8.

The algorithm draws N intervention configurations under shared noise. The active suspect set $\mathbf{C}_j$ (line 10) is drawn at uniformly chosen cardinality $k_s \in \{1, \dots, n_s\}$; the dropped witness set D_j (line 11) is drawn the same way at $k_w \in \{0, \dots, n_w\}$, and the active witnesses on line 12 are the complement $\mathbf{W} \setminus D_j$, so n_w bounds the number of witnesses *dropped*, not the number kept, and the active witness set has size $|\mathbf{W}| - n_w$ or larger. The witness ablation flag `wit = off` is handled once on line 2 by emptying $\mathbf{W}$; this collapses the witness machinery on lines 11–12 and 14–15 to no-ops, producing the dotted accuracy curves of Figure 8.

Alternative values on line 13 are bit flips: on $\{0, 1\}$ the ϵ -excised alternative-value distribution has a single support point and ϵ plays no role; the explicit sampling form is preserved so that the same routine runs on continuous benchmarks. The two interventional worlds on lines 14–15 share the noise realisation $\mathbf{u}_j$ drawn on line 8, the standard counterfactual coupling, which lets the per-suspect score on lines 16–18 be read off as the nan-mean of $|y_j^n - y^*|$ over the draws in which $X \in \mathbf{C}_j$. The sufficiency outcome y_j^s is computed but not consumed: the AC verdict against which we benchmark has no sufficiency clause, so we read only the necessity world. The joint estimator with sufficiency active is exercised in Sections 5, 6, and 10, where AC has no comparable verdict to benchmark against. At each site $i = 1, \dots, n$, line 24 returns the suspect, A_i or B_i , with the higher mean.

Benchmark protocol. For each problem size n we sample up to ten distinct factive worlds (with $Y = 1$), instantiate the model, run both methods to verdicts, and compare to the analytic ground truth. Each method is granted a 60-minute global budget per run; if the budget is exhausted mid-loop, the largest completed n caps the curve. The approximate estimator is exercised in five configurations

Algorithm 1 Approximate PCI estimator on the scaled throwing problem.

Require: PSCM M ; factual world $(\mathbf{s}^*, \mathbf{w}^*, y^*)$; cardinality bounds n_s, n_w ; sample budget N ; witness flag $\text{wit} \in \{\text{on}, \text{off}\}$
Ensure: per-site verdicts $v_i \in \{A_i, B_i\}$ for $i = 1, \dots, n$

```
1: if  $\text{wit} = \text{off}$  then
2:    $\mathbf{W} \leftarrow \emptyset$  ▷ no witness candidates
3: end if
4: for each  $X \in \mathbf{S}$  do
5:    $L[X] \leftarrow []$  ▷ per-suspect score list
6: end for
7: for  $j = 1, \dots, N$  do
8:   draw  $\mathbf{u}_j \sim P_{\mathbf{U}}$ 
9:   sample  $k_s \sim \text{Unif}\{1, \dots, n_s\}$ ,  $k_w \sim \text{Unif}\{0, \dots, n_w\}$ 
10:  sample  $\mathbf{C}_j \subseteq \mathbf{S}$  uniformly with  $|\mathbf{C}_j| = k_s$ 
11:  sample  $D_j \subseteq \mathbf{W}$  uniformly with  $|D_j| = k_w$  ▷ witnesses dropped
12:   $\mathbf{T}_j \leftarrow \mathbf{W} \setminus D_j$  ▷ active witnesses
13:   $\mathbf{c}'_j \leftarrow \mathbf{1} - \mathbf{s}^*|_{\mathbf{C}_j}$  ▷ bit-flip on  $\{0, 1\}$ 
14:   $y_j^s \leftarrow M.\text{run}(\mathbf{u}_j, \text{do}(\mathbf{C}_j = \mathbf{s}^*|_{\mathbf{C}_j}, \mathbf{T}_j = \mathbf{w}^*|_{\mathbf{T}_j}))$  ▷ sufficiency
15:   $y_j^n \leftarrow M.\text{run}(\mathbf{u}_j, \text{do}(\mathbf{C}_j = \mathbf{c}'_j, \mathbf{T}_j = \mathbf{w}^*|_{\mathbf{T}_j}))$  ▷ necessity, shared  $\mathbf{u}_j$ 
16:  for each  $X \in \mathbf{C}_j$  do
17:    append  $|y_j^n - y^*|$  to  $L[X]$ 
18:  end for
19: end for
20: for each  $X \in \mathbf{S}$  do
21:    $\text{score}[X] \leftarrow \text{nanmean}(L[X])$ 
22: end for
23: for  $i = 1, \dots, n$  do
24:    $v_i \leftarrow \arg \max(\text{score}[A_i], \text{score}[B_i])$ 
25: end for
26: return  $(v_1, \dots, v_n)$ 
```

that vary along three axes identified in Section 3: (i) initial sample budget $N_0 \in \{250, 500\}$, scaled linearly with n by a factor $\alpha \in \{0.5, 1\}$ so that the total sample at size n is $\alpha \cdot N_0 \cdot n$; (ii) presence of witness interventions ($\text{wit} \in \{\text{on}, \text{off}\}$), ablating $\mathbf{T} \leftarrow \mathbf{w}^*$ to test whether witness pinning is load-bearing; (iii) cardinality bounds n_s, n_w , fixed at 4 in the default runs (answering “does some witness set of size $\geq |\mathbf{W}| - 4$ work?”) and grown linearly as $4n$ in the dynamic-set-size configuration (recovering the unrestricted subset search at the cost of higher variance). The cardinality bound is a tunable approximation knob, not a hidden assumption: the dynamic-set-size run measures the empirical cost of bounding n_s, n_w at 4 on this problem, and the gap between the two configurations is what the ablation surfaces. The conceptual case for constraining cardinality at all, that uniform sampling over the full powerset biases attributions toward large-coalition interventions and is analogous to declining to enumerate all n -way regression interaction terms, is given in Section 3.

Results. Three observations summarise the experiment. First, runtime (Figure 6): the exact procedure times out at $4n + 1 = 17$ variables ($n = 4$) once the cause-witness search space crosses sixty-five thousand pairs, while the approximate methods continue to roughly 73–145 variables depending on sample budget, with the frugal $\alpha=0.5$ run reaching the largest sizes. Second, search-space size (Figure 7, log scale): the exact method visits the full powerset, while the approximate methods visit at most a few hundred unique configurations per problem size: the gap is several orders of magnitude and widens with n . Third, accuracy (Figure 8): the exact method is correct by construction up to its early demise; the witness-using approximate methods stay above 0.85 correct-attribution rate up to roughly 60 variables and degrade gracefully beyond that, with the dynamic-set-size run sustaining among the highest accuracies at moderate problem sizes. Witness-free runs (dotted) underperform throughout, starting near 0.67 already at $n = 1$. The ablation confirms a structural claim of Section 3: pinning factual witness values is doing real work, not just regularising.

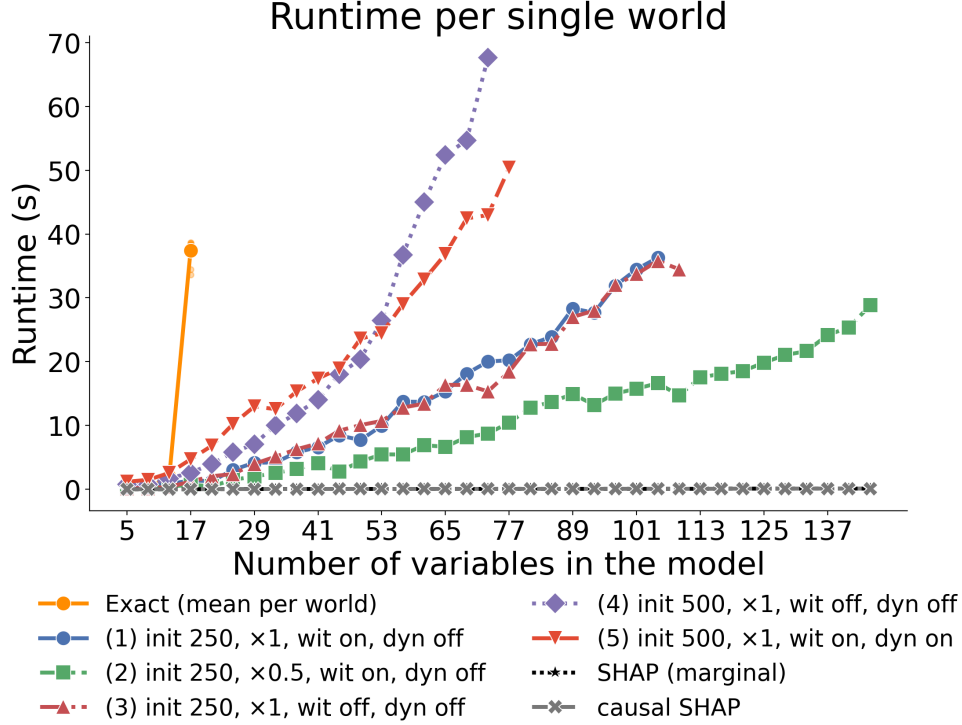


Figure 6: Per-world runtime against problem size. The exact method (orange) times out at 17 variables; approximate methods continue, with reach controlled by total sample budget. Five approximate configurations vary initial sample size, the linear-growth factor α , the witness ablation, and dynamic cardinality bounds; see Benchmark protocol.

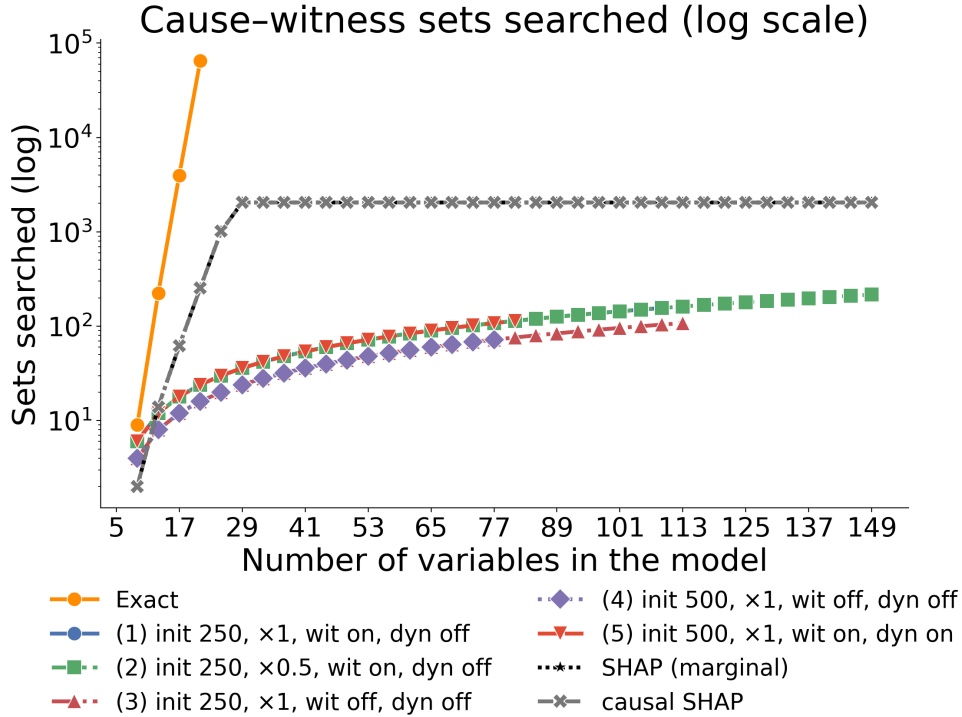


Figure 7: Number of cause-witness configurations explored per problem size, log scale. The exact method visits the full powerset; the approximate methods visit a few hundred unique configurations per problem size regardless of n , a gap of several orders of magnitude that grows with n .

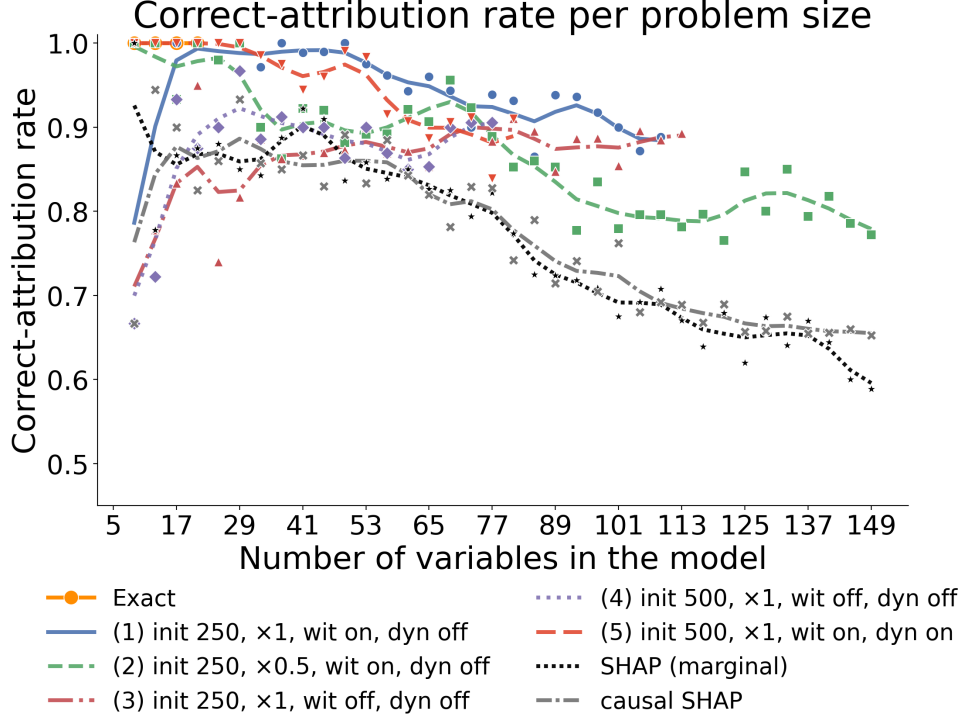


Figure 8: Correct-attribution rate against problem size. Each approximate configuration is lightly smoothed (Gaussian filter, $\sigma = 1$) with its raw per-size points overlaid, and is shape-coded in the legend (marker, colour, and line style). The exact method is correct up to 17 variables. Configurations using witnesses stay above 0.85 up to roughly 60 variables and degrade gracefully thereafter; the witness-free configurations (wit off) underperform throughout, starting near 0.67. Dynamic-set-size sampling (dyn on) is among the highest at moderate problem sizes.

Scope of the conclusion. The benchmark establishes that the necessity-only PCI estimator tracks AC verdicts on a faithful scaling of the canonical example with several orders of magnitude fewer evaluated configurations. The problem itself, however, is structured: responsibility attribution is independent across sites, exactly one of A_i, B_i is responsible at each site given $C=1$, supersets of any witness set witness the same attribution (so a small witness-cardinality bound costs little), and all variables are binary. These regularities account for some of the favourable scaling: in particular, the cardinality-bounded sampling losing little to dynamic-set-sizes sampling on this problem. Two extensions are natural follow-ups. First, a comparison against SHAP and causal SHAP on the same scaled-throwing problem is deferred to a companion benchmark (cf. Section 5 for the corresponding comparison on the continuous overdetermination example); the sufficiency component of PCI is what distinguishes it from those baselines, and a clean head-to-head requires the joint kernel rather than the necessity-only restriction used here. Second, benchmarks that break each of the structural regularities above, particularly continuous-variable extensions where AC’s mechanical enumeration is no longer even well-defined, would test the estimator outside its current favourable regime. As a sanity check that the correspondence in Theorems 28–29 is empirically operational at scale, the present comparison is sufficient.

Where we go next. The AC benchmark exercises the necessity-only specialisation of PCI on a discrete-outcome problem with a closed-form ground truth. Section 10 shifts to the regime where AC has no verdict to compare against: a Bayesian dynamical SIR model with a continuous outcome and asymmetrically interacting policies. The point is to exercise the joint necessity-sufficiency machinery on threshold events that the actual-causality machinery cannot adjudicate, and to show that PCI separates lockdown and masking where but-for analysis collapses them.

10 Dynamical SIR Benchmark: PCI on Continuous Outcomes

The actual-causality benchmark of Section 9 exercises the necessity-only specialisation of PCI on a discrete outcome with a closed-form ground truth. This section shifts to the regime where AC has no verdict to compare against: a Bayesian dynamical model with a continuous outcome and asymmetrically interacting causes. The benchmark mirrors a public ChiRho tutorial²³ in setup and query, which uses ChiRho’s `SearchForExplanation` handler to compute classical actual-cause probabilities. We keep the same model and the same query but replace the explanatory machinery with the PCI thin-search sampler. The full implementation, raw outputs and plotting code are in the companion notebook `docs/source/sir_benchmark.ipynb`.

Model. The dynamics are the standard SIR system $\dot{S} = -\beta SI$, $\dot{I} = \beta SI - \gamma I$, $\dot{R} = \gamma I$, with Bayesian priors $\beta \sim \text{Beta}(18, 600)$ and $\gamma \sim \text{Beta}(1600, 1600)$. Two non-pharmaceutical policies, each enacted with prior probability $1/2$, modulate the transmission rate via an intervention strength $l \in [0, 1]$ that scales β_0 to $(1-l)\beta_0$. Lockdown alone has efficiency 0.6; masking efficiency depends on lockdown (0.1 under lockdown, 0.45 on its own); the joint efficiency is the clamped sum, capped at 0.95. Lockdown is enacted at $t = 1$, masking at $t = 1.5$. The outcome is the *overshoot*, the peak-to-final S gap of the simulated trajectory, and the undesirable event is $\text{os_too_high} = \mathbb{1}[\text{overshoot} > 24]$ on a 100-person population.

But-for analysis fails to separate the suspects. Conditioning on each pair of policy decisions and drawing 100 predictives yields $\Pr(\text{overshoot} > 24) \approx 0.07$ with no intervention, 0.67 with both policies, but 0.82 with lockdown only and 0.87 with mask only — so, since a higher overshoot probability marks a *weaker* policy, mask-only is the weakest of the three intervention regimes. The mechanism is the asymmetric joint efficiency: combining the two policies caps the transmission reduction at 0.95, but neither policy alone reaches that cap, and mask-only is comparatively weak unless lockdown is absent (which shifts mask efficiency from 0.1 to 0.45). But-for cannot disentangle this context dependence: it treats each policy as a binary cause and reports an aggregate effect, with no way to express that lockdown’s causal contribution depends on whether masking is in force. Figure 9 visualises the four interventional regimes: the trajectory bands collapse toward the un-intervened baseline only under both policies, while either policy on its own leaves the infectious peak essentially intact, mirroring the near-equal but-for overshoot probabilities.

What a correct diagnosis should say. The but-for numbers are symmetric, but the model is not, and the asymmetry points to a clear qualitative reading that any responsibility method ought to recover. First, *lockdown is the proximate cause of excessive overshoot*: it is the policy that does the real work of suppressing the infectious peak, and the joint regime clears the 24-overshoot threshold only because lockdown is in force. Second, *masking is at most a context-sensitive cause*: its contribution depends on whether lockdown is also active. Masking is comparatively effective only when lockdown is absent (efficiency 0.45 versus 0.1 under lockdown), so once lockdown is present masking adds little. A good attribution method should therefore rank lockdown above mask and, more finely, flag mask’s causal status as contingent on context rather than robust. These two intuitions are the target of the analysis below; they follow from the model’s asymmetric joint efficiency, not from any particular tool, and they coincide with the verdict the ChiRho tutorial reaches via classical actual-cause probabilities. We state them here so the comparison that follows does not depend on consulting that tutorial.

PCI thin-search procedure. The suspects are $\{\text{lockdown}, \text{mask}\}$. The witnesses are drawn from the policy efficiencies $\{\text{lockdown_efficiency}, \text{mask_efficiency}, \text{joint_efficiency}\}$, with the suspect itself excluded from the witness set on each draw. The factual world fixes both policies on ($\text{lockdown} = 1, \text{mask} = 1$), so the factual outcome $y^* = \text{overshoot}$ is read from the same trajectory. For each suspect the sampler draws 2,500 regimes, scoring each with the $|y^* - y_{\text{nec}}| - |y^* - y_{\text{suff}}|$ statistic and averaging across regimes. The *nec* term rewards regimes where removing the suspect moves the outcome *away* from y^* ; the *suff* term rewards regimes where fixing the suspect at factual keeps the outcome *close* to y^* .

The joint necessity-sufficiency plane. Before turning to numbers, it helps to see what the search produces. For each suspect the sampler returns, per regime, a pair of outcomes: the *necessity*-

²³https://basisresearch.github.io/chirho/explainable_sir.html

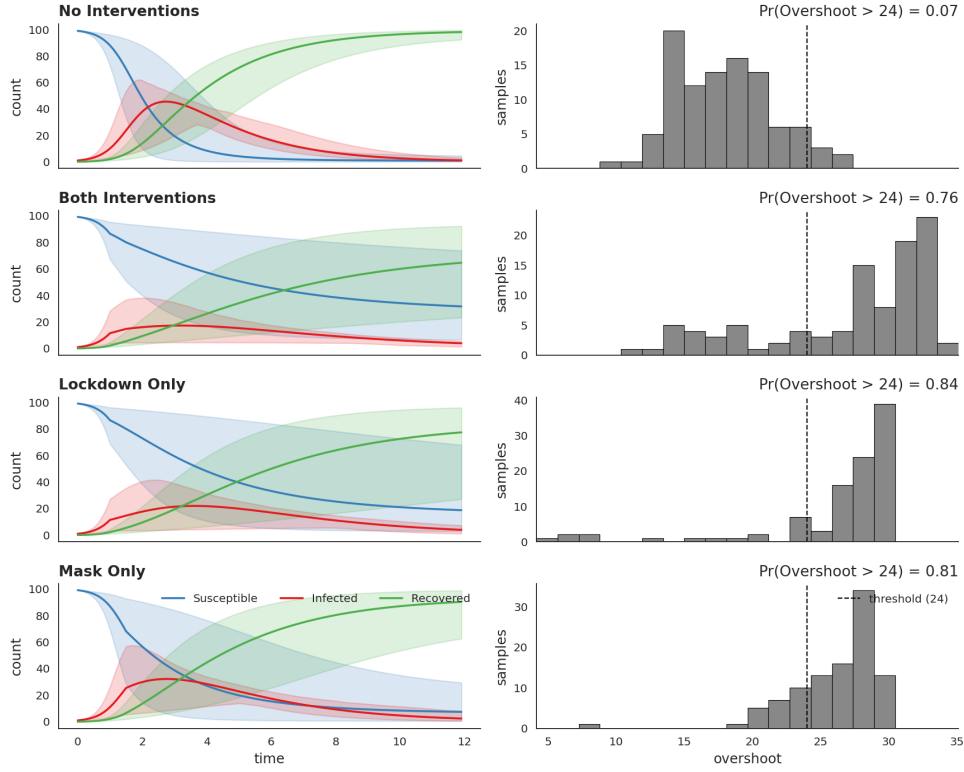


Figure 9: SIR trajectories and overshoot distributions under the four interventional regimes (no interventions, both policies, lockdown only, mask only). Left: posterior bands for susceptible (blue), infectious (red) and recovered (green) over time, with the lockdown enacted at $t = 1$ and the mask mandate at $t = 1.5$. Right: per-regime histograms of the overshoot, with the 24-overshoot threshold marked. Either policy on its own leaves the infectious peak essentially intact; only the joint regime suppresses it appreciably, yet the lockdown-only and mask-only overshoot probabilities are near-indistinguishable: the asymmetric joint efficiency that PCI disentangles.

world overshoot y_{nec} (the focal suspect set to its alternative, any other active suspects in $\mathbf{C} \setminus \{X\}$ intervened, witnesses pinned at factual) and the *sufficiency-world* overshoot y_{suff} (the focal suspect held at factual, other active suspects intervened, witnesses at factual). These are the necessity- and sufficiency-world outcomes y^n, y^s of Section 3; we write $y_{\text{nec}}, y_{\text{suff}}$ here to match the figure axes. Plotting one point per regime in the $(y_{\text{nec}}, y_{\text{suff}})$ plane gives a per-suspect cloud (Figure 10) that exposes the score geometry before any averaging. Three reference lines orient the plane: the dotted red cross-hair marks the factual outcome y^* at the both-policies-on world (lockdown = mask = 1); because that world is the high-overshoot scenario, y^* sits near the upper-right of the cloud rather than at its centre. The dash-dot blue line marks the 24-overshoot threshold, and the dashed black diagonal is $y_{\text{nec}} = y_{\text{suff}}$. The per-regime score $|y^* - y_{\text{nec}}| - |y^* - y_{\text{suff}}|$ is positive exactly when a point lies *further from the cross-hair along the necessity axis than along the sufficiency axis*: necessity pulls the outcome away from factual (evidence the suspect was needed), while sufficiency keeps it close (evidence the suspect alone sustains the outcome). A cause therefore shows up as a cloud displaced toward smaller y_{nec} at fixed y_{suff} — the *cause signature* on this geometry — whereas a non-cause hugs the diagonal. Reading the two panels this way, the lockdown cloud is clearly displaced below the diagonal toward smaller y_{nec} , while the mask cloud sits closer to it: the visual form of the verdict the next paragraph quantifies.

Single-world results. PCI assigns lockdown roughly $2.0\times$ the responsibility of mask:

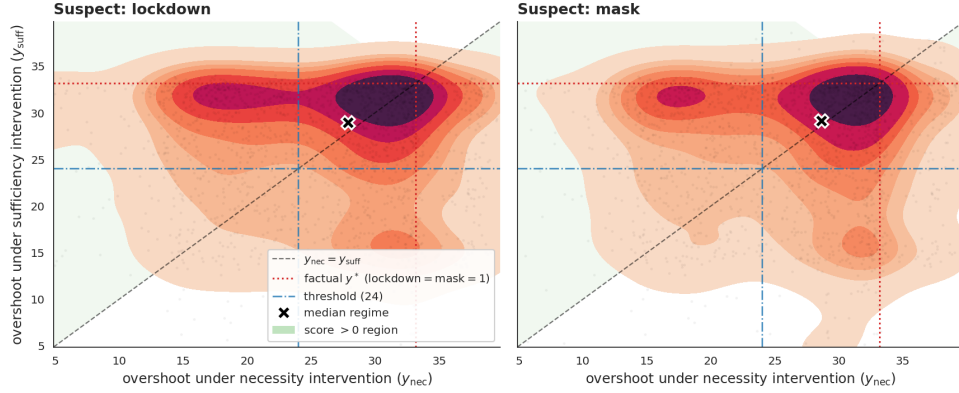


Figure 10: Joint necessity-sufficiency overshoot density per suspect. Each point is one sampled regime. Dotted red cross-hair: factual y^* at lockdown = mask = 1 (the high-overshoot anchor); dash-dot blue: the 24-overshoot threshold; dashed black: $y_{nec} = y_{suff}$. A regime contributes positively to the PCI total iff it sits further from the red cross-hair along the necessity axis than along the sufficiency axis. The lockdown density is shifted toward smaller y_{nec} relative to y_{suff} (the cause signature on this geometry); the mask density sits closer to the diagonal.

suspect	$ y^* - y_{nec} $	$- y^* - y_{suff} $	total
lockdown	8.12	-6.58	1.54
mask	7.43	-6.67	0.76

The gap is driven entirely by the necessity term: removing lockdown moves the overshoot further from y^* than removing mask does (8.12 vs 7.43 in mean absolute distance), while the sufficiency terms agree to within 0.1. This is consistent with the but-for finding that either policy *alone* is roughly equally bad. What distinguishes the suspects is that lockdown’s removal *under context* (with witnesses pinned to factual) pulls the world further from y^* than mask’s removal does — the lockdown cloud’s displacement below the diagonal toward smaller y_{nec} in Figure 10, against the mask cloud’s near-diagonal position.

Robustness across factual worlds. We replicate the analysis across 10 factual worlds drawn from the prior (rather than the single hand-set world above), running thin search on each and averaging. Lockdown’s mean total exceeds mask’s in the majority of worlds; the verdict is not an artifact of the chosen factual.

Effect of context. The ChiRho tutorial isolates one downstream variable at a time, asking whether the verdict survives when that variable is held fixed at factual versus left free. We replicate the experiment by partitioning the PCI regimes by whether the partner-policy efficiency variable was sampled into the witness set (Figure 11). For lockdown, with mask_efficiency *fixed* as a witness the score is 0.81; *free*, it rises to 2.67. For mask, with lockdown_efficiency *fixed* the score is -0.48 (mask fails to register as a cause at all under that context regime) and *free* it rises to 2.72. This experiment partitions the regimes differently from the single hand-set world reported earlier, so the fixed/free scores are not expected to average to that world’s totals (lockdown 1.54, mask 0.76). Lockdown therefore registers as a cause in both context regimes, while mask’s status as a cause is contingent on its partner being allowed to adapt. PCI thus expresses not just *which* policy is the dominant cause but *how robustly* that causal claim holds across counterfactual contexts: a finer-grained verdict than the binary “actual cause / not” that classical AC produces.

Takeaway. On the dynamical SIR setting, PCI recovers both intuitions set out at the start of this section — lockdown is the proximate cause of excessive overshoot and masking only a context-sensitive one, despite both policies appearing similar under but-for analysis — and expresses them on a continuous, decomposable scale. The necessity-vs-sufficiency decomposition distinguishes suspects whose isolation effects are similar but whose removal-under-context effects diverge; the

Effect Of Fixing The Partner-Policy Efficiency As Witness

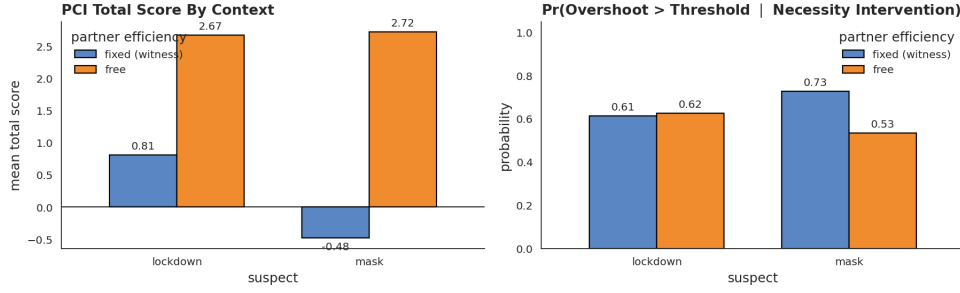


Figure 11: Different-contexts experiment. Bars compare PCI total score (left) and $\Pr(\text{overshoot} > 24 \mid \text{necessity})$ (right) when the partner-policy efficiency variable is fixed as a witness versus left free. Lockdown’s score is positive in both regimes; mask’s score flips sign across regimes, indicating that mask is a context-sensitive rather than a robust cause.

contexts experiment further distinguishes robust from context-sensitive causes. The same machinery applies to any ChiRho probabilistic program with a continuous outcome and a tractable enumeration over interventional regimes.

Where we go next. The SIR benchmark scales PCI to a research-grade Bayesian dynamical model and demonstrates that the joint kernel captures patterns that AC cannot adjudicate. Section 11 pushes the demonstration one step further: a deployed automated valuation model trained on millions of points, with deep neural-network components and a sparse variational GP outcome head. The question there is no longer whether PCI agrees with a known ground truth (on a proprietary production model none is available), but whether the computation is feasible at all, and whether the qualitative SHAP-vs-PCI divergences observed on the synthetic side persist on a real trained system.

11 Scaling PCI to a Real-World Automated Valuation Model

Where we are. The synthetic archetypes of Section 5, the AC benchmark of Section 9, and the SIR benchmark of Section 10 have all controlled the model and the ground truth, so that PCI could be judged against either an analytic verdict or a competing AC enumeration. This section drops both: we apply PCI to a deployed production model on which neither ground truth nor exact AC enumeration is available, and ask whether the computation remains feasible and whether the qualitative SHAP-vs-PCI divergence persists.

In this section, we apply PCI to a valuation model for residential housing. Disclosure constraints on the trained model and its training data prevent us from publishing a quantitative accuracy comparison against ground truth on this system; the quantitative validation of PCI therefore lives entirely in the controlled synthetic benchmark of Section 5, against an analytic ground truth that the production model does not admit. The role of this section is correspondingly narrow: to report (i) that PCI is computationally feasible at production scale on a model with millions of training points, and (ii) that the qualitative SHAP-vs-PCI divergence observed in Section 4 persists when the model is a real trained one. What is reportable here is the shape of the two attribution distributions and the structural pattern of where attribution mass lands, not the substantive content of the features they assign to.

Setting. The unit of analysis is a residential property transaction. PCI is run against the full country-level production AVM; what we show in this section, however, is an illustrative prototype based on the state of New York rather than the full national model structure, which we cannot disclose in detail. The NY prototype shares the architecture and the causal-kernel construction described below with the full system but is smaller in scope and DAG complexity. The computational-feasibility claim refers to the full national system; the qualitative SHAP-vs-PCI behaviour we illustrate is a property of the shared architecture and carries over from the prototype. The outcome

variable y is the (log-epsilon-standardized) sale price. Transactions are described by a set of categorical variables $x_{cat} = x_1, \dots, x_k$ and continuous (transformed) variables²⁴ $x_{con} = x_{k+1}, \dots, x_n$.

Causal model. A causal DAG $\mathcal{G}$ expresses our causal assumptions about the relationships between variables. It was written manually in collaboration with domain experts and refined by adding edges to break implied conditional independencies that are not upheld by the data. The NY-prototype DAG has 25 nodes and 99 edges; topology with anonymised node labels is shown in Figure 12; the full national DAG is larger and is not shown. As demonstrated in Section 9, even the prototype is already large enough that exact actual-causality enumeration is computationally infeasible, and the same conclusion applies a fortiori to the national system.

Variables x_i are drawn from distributions Dist_i parameterized by functions $f_i(\text{pa}_i, \varphi_i)$ of the parent variables pa_i ²⁵: $x_i \sim \text{Dist}_i(f_i(\text{pa}_i, \varphi_i))$. We call these distributions *causal kernels*. The distributions are either (truncated) normal or categorical, chosen based on the domain of the variable that they model. The functions f_i are deep neural networks with parameters φ_i . The parameters φ_i are optimized using cross-entropy loss for categorical distributions and average negative log likelihood for continuous distributions. The outcome is modeled by a sparse variational Gaussian Process (SVGP; Titsias, 2009, Hensman et al., 2013), $x_{gp} = (x_{con}, \eta(x_{cat}, \varphi_{cat}))$, $y \sim \text{SVGP}(\mu(x_{gp}, \varphi_\mu), \kappa(x_{gp}, \varphi_\kappa))$, where η is a linear embedding of the categorical variables into the continuous space, jointly optimized with the GP.

Feasibility at production scale. To apply PCI we transform the model as described in Section 3 and evaluate the impact of features from a list of potential causal suspects, using the absolute difference impact score of Example 21. We draw Monte Carlo samples from the expanded model. Conditioning on a given suspect X_i being active gives us a sample of the scores corresponding to that feature, which we use to estimate the expected value thereof. The estimation converges at around 25,000 samples; for a batch of 50 data points this takes around 60 minutes on commodity hardware. Sampling needs to be performed only once; the samples can then be reused to compute the attribution score of any suspect. This establishes that the method is computationally tractable at production scale, not a foregone conclusion, since the expanded-model construction in Section 3 introduces additional latent dimensions per suspect and per witness.

Qualitative comparison to SHAP. We compare PCI attributions to SHAP scores [Lundberg and Lee, 2017] on the same trained model. SHAP heavily weights variables that are downstream of many other variables in the causal graph, effectively assigning attribution to what amount to summary statistics of the outcome. PCI distributes attribution more evenly across the upstream causal structure. The pattern is consistent with what Section 4 establishes on controlled examples: SHAP’s observational marginalisation allocates weight differently from intervention-and-witness-based PCI scores, and on a deep DAG this difference manifests as concentration on summary-statistic-like nodes. Figure 13 shows the two attribution distributions with anonymised feature labels; what is reportable here is the shape of the two distributions, not which features they assign to.

Where we go next. Section 12 closes the paper by articulating the relationship between PCI and the two traditions it inherits from, Halpern’s actual causality and Pearl’s PN/PS/PNS, against the background of what every preceding section has shown, and points out the limitations of the present construction along with the natural follow-up directions.

12 Discussion and Conclusions

We are now in a position to articulate the main differences between PCI and the two approaches that inspired it, against the background of what the rest of the paper has shown. The main lesson from the literature on actual causality is that intuitive causal attribution often requires holding some endogenous variables, members of a “witness set”, fixed at their factual values. This ensures context sensitivity by controlling complicating causal paths. We treat witnesses somewhat differently: instead of looking for a single witness set that reveals a causal relationship, we use a distribution

²⁴Usually log-epsilon-standardized, standardized-truncated, or standardized.

²⁵For root nodes, pa_i is empty.

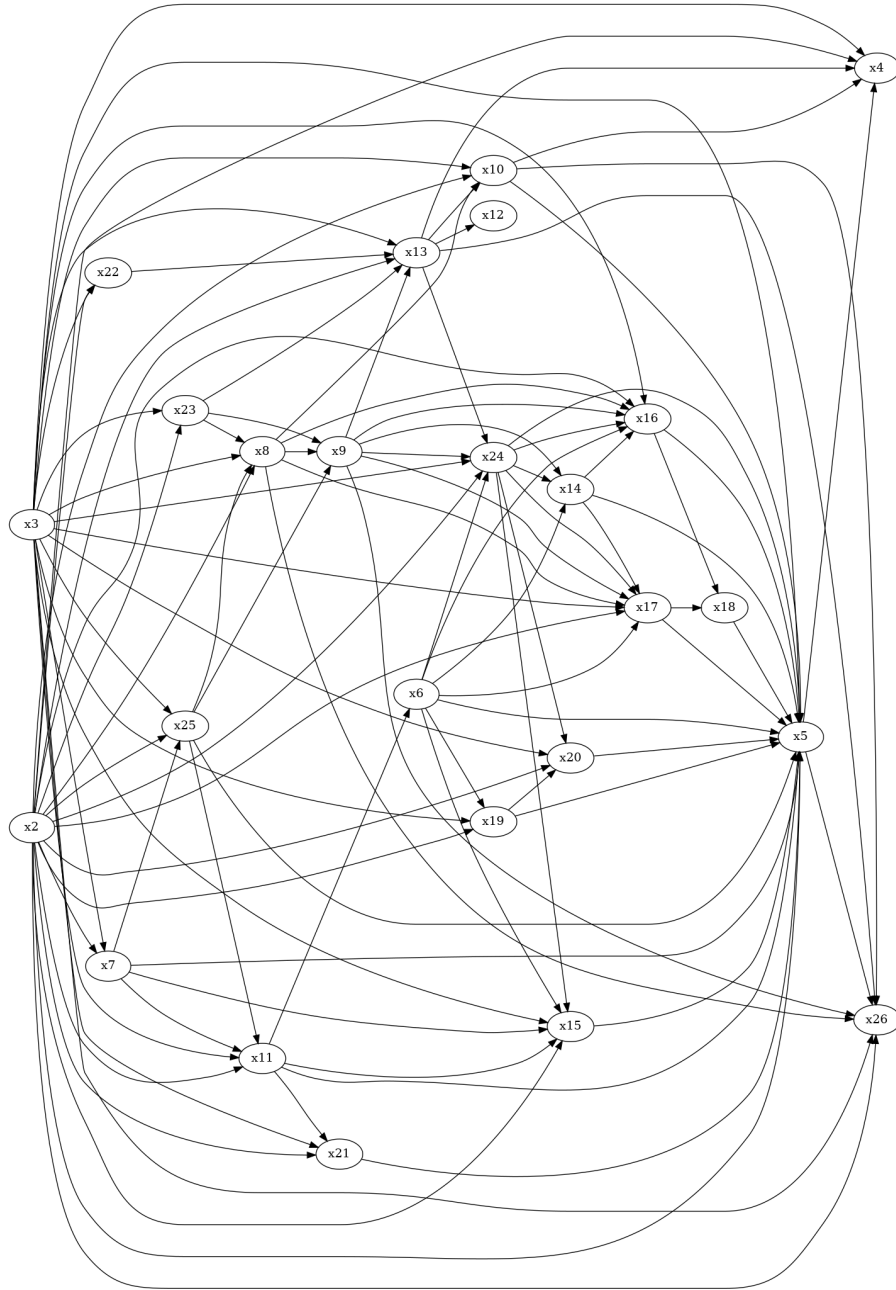


Figure 12: Topology of the AVM causal DAG (NY-prototype version), with node labels anonymised. The graph has 25 nodes and 99 edges; the full national DAG is larger and is not shown. Several variables sit deep in the DAG with high in-degree; these are the nodes towards which SHAP allocates disproportionate attribution mass (see Figure 13).

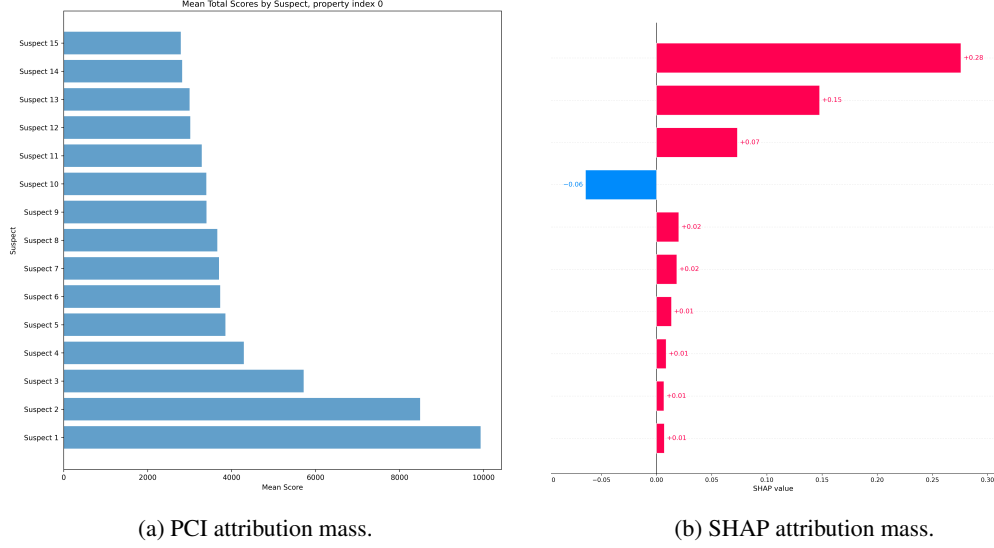


Figure 13: Attribution distributions on the AVM, with feature labels anonymised. **Left:** PCI attribution mass spread across upstream causal variables (suspects 1–15, ranked by mean total score). **Right:** SHAP attribution mass concentrated on a small number of downstream variables. The qualitative pattern matches the controlled-setting comparison of Section 4.

over witness sets and integrate the causal impact score across them. As in many probabilistic generalisations, this focuses on distributions rather than point estimates, and it lets PCI account for the number of witness sets and the strength of within-witness-set effects. Moreover, the construction admits off-the-shelf probabilistic inference rather than worst-case subset enumeration. The relationship between the two views is now formal rather than gestural: under matching choices of Γ and Δ , AC verdicts coincide with the support of $\sum_{\mathbf{T}} \Phi$ (Theorems 28–29 and Proposition 34), and the empirical companion in Section 9 confirms this at scales where AC enumeration is intractable.

The main concept borrowed from Pearl’s probability of necessity and sufficiency is the joint distribution over outcomes from two interventional regimes, pushed forward through a user-chosen causal impact function. The key difference is that we use witness sets to allow for context-sensitivity and to break symmetries present in otherwise pathological cases like overdetermination and undercutting. Section 4 works through the binary instantiation in which the necessity and sufficiency components coincide with PN and PS, and Section 5 reads the two components apart on a continuous synthetic model where each is needed to surface a different causal archetype.

The further generalisations are mostly bookkeeping in the same spirit. Instead of fixing a single alternative value for a potential cause, we integrate over a factual-excised alternative-value distribution; this generalises immediately to continuous variables. Instead of asking whether a particular set is an actual cause, we sample suspect sets and gauge a feature’s responsibility in terms of its participation in sets with causal impact. Instead of asking whether the outcome differs, a customisable causal impact function ci measures the differences across counterfactual worlds, including between continuous variables. The variable-selection distribution Γ , the alternative-value distribution Δ , and the impact function ci become explicit, user-facing components of the framework rather than implicit defaults.

Against SHAP-family attributions, the framework’s central commitment is to *intervention with witnesses* rather than *observational marginalisation*. The controlled comparisons of Section 4 show that this commitment is load-bearing on canonical overdetermination and continuous-mediation examples, where SHAP and Causal SHAP fail in characteristic ways that PCI does not. The synthetic benchmark of Section 5 confirms this on a model with known ground truth, and the case study of Section 11 reports the same qualitative pattern on a production-grade trained model with millions of data points. A separate comparison to gradient-based attribution, where the contrast is differentiability and locality rather than the causal/non-causal axis, is reported in Section 6. The thread connecting these results is structural: the witness mechanism that satisfies D-A-rank for Alice in

the binary OBCB walkthrough of Section 4 is the same machinery driving the upstream-versus-downstream attribution split on the AVM in Section 11.

A broader test is how PCI fares against the desiderata that the XAI literature typically asks counterfactual-explanation methods to meet, of which Verma et al. [2024] list five. *Sparsity*, the requirement that explanations change as few features as possible, follows directly from the cardinality cap on suspect sets in Γ , which biases the search toward small C . *Multiple counterfactuals* (the ability to return several alternative explanations rather than commit to one) falls out of the Monte Carlo construction, since a single sweep over Γ and Δ produces a population of (C, c', T) triples rather than a single optimum. *Causal-constraint satisfaction*, the demand that explanations respect downstream propagation rather than treat features as independently editable, is automatic once suspect interventions are routed through the causal model: changing C propagates to descendants by construction. *Model-agnosticism* (applicability to arbitrary predictors) PCI satisfies only partially: it can wrap a black-box predictor, but at the cost of the causal sensitivity that motivates the framework in the first place.

Actionability, the requirement that explanations not recommend changes to features users cannot in practice change, is where we part company with the literature. The standard recommendation is to fold ease-of-change into the cause-identification procedure itself, weighting features by mutability so that immutable ones are never returned as causes (Verma et al. [2024]; cf. Karimi et al. [2021], who shift the focus from explanations altogether to interventions). We think this conflates two distinct questions. A procedure that suppresses gender as a possible cause whenever a loan decision is at stake will help applicants game the existing system, but it will also prevent them from noticing the kind of structural problem that responsibility attribution can in principle surface. Cost-of-action layers can sit downstream of PCI for users who need actionable recommendations, but they should compose with responsibility attribution rather than replace it.

Several limitations are worth flagging. PCI presumes access to a trained probabilistic causal model from which counterfactual outcomes can be sampled, a stronger requirement than methods that work from observational data and a causal graph alone. The Monte Carlo variance of the estimator grows with model size and with the cardinalities of suspect and witness sets; Section 9 reports empirical scaling and an ablation that isolates the contribution of witness pinning, but sample-efficient estimators for high-cardinality settings remain an open direction. Finally, the AVM case study is deliberately limited in disclosure: the apples-to-apples SHAP-vs-PCI comparison rests on the synthetic benchmark of Section 5, with the production model serving as a feasibility witness rather than as quantitative evidence.

Where PCI falls short. Three caveats about PCI’s reach deserve explicit discussion. (i) *AC mismatch under noise integration.* Observation 33 shows that integrating the kernel against P_U before maximisation can drop AC causes that are subset-minimal but not cardinality-minimal: the Γ -weighted ranking is governed by cardinality, while AC’s minimality is governed by the subset relation. Proposition 34 provides the operational route to AC verdicts in such cases; Theorem 31 closes the gap only under *necessity dilution* (Definition 30), a regularity that fails in disjunctive overdetermination. (ii) *Cardinality knob versus sample efficiency.* The cardinality cap is a tunable hyperparameter rather than a hidden default, but it trades precision for variance: tight bounds shrink the search space and reduce Monte Carlo variance per iteration but exclude joint-cause sets that AC’s exhaustive enumeration would surface; loose bounds restore generality at the cost of slower convergence. Problem families with diffuse joint causation need more generous settings than the canonical AC benchmark uses, and the cap is therefore worth reporting alongside results. (iii) *Cost of the PSCM-availability assumption.* Pointwise verdicts at the configuration level require a sampleable structural model. Methods that bound PN/PS/PNS from observational data and a causal graph alone, e.g. Tian and Pearl’s bounds, apply in settings where PCI cannot, and the stronger PSCM commitment should be weighed against this when only a graph and a dataset are available.

Taken together, PCI provides a scalable, context-sensitive, causal explanation method for causal probabilistic machine-learning models. A natural probabilistic generalisation of ideas already in the literature leads to a tractable estimator that can be paired with standard inference algorithms. We benchmarked the approach against exact actual-causality computations, characterised its qualitative behaviour on synthetic models with known ground truth and on canonical SHAP failure cases, and showed its computational feasibility on a production-grade model.

Responsibility attribution has not yet been effectively mechanised in software for large, complicated models, which makes the exact notions present in the literature infeasible to apply. PCI develops approximate causal attribution workable for users of models of non-trivial size, with the design choices — which variables are suspects, what counts as an alternative value, what counts as a difference — exposed to the user rather than baked in.

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A Computations for the OBCB Running Example

This appendix collects the worked computations for the stochastic OBCB model of Example 6 that are summarised in the main text: the per-branch traces behind Alice’s individual PN and PS values in Table 1 (Section 2), and the closed-form SHAP value function behind Tables 4–5 (Section 4.2). The remaining individual and population entries of Table 1 are obtained analogously; all values are reproduced in `docs/source/obcb_computations.ipynb`.

A.1 Alice’s probability of necessity

Alice is factually $gender = \text{female}$, $credit = \text{bad}$, $loan = F$; the counterfactual intervenes to set $gender = \text{male}$ while leaving $credit = \text{bad}$. Tracing through the model:

- Under the intervention, $check \sim \text{Bern}(0.9)$: men are checked 90% of the time.
- If checked, $check\text{-}failed = 1$ (Alice’s credit is bad), and from the loan-probability table $P(loan\text{-}if\text{-}checked = 1 \mid gender = \text{male}, check\text{-}failed = 1) = 0.05$.

- If not checked, $loan = 0$ deterministically since $loan = loan\text{-if-checked} \cdot check$.

Combining the two branches:

$$\begin{aligned}
PN(gender = \text{female}, loan = F \mid \text{Alice}) \\
&= P(loan_{gender=\text{male}} = T \mid gender = \text{female}, credit = \text{bad}) \\
&= 0.9 \cdot 0.05 = 0.045.
\end{aligned}$$

A.2 Alice’s probability of sufficiency

Following the same convention as for PN , the individual-level computation additionally conditions on Alice’s credit ($credit = \text{bad}$). The PS conditional then has Alice’s credit fixed but the cause and outcome flipped to their non-factual values ($gender = \text{male}$, $loan = T$); we intervene to restore $gender = \text{female}$ and ask whether the loan is again denied:

- With $gender = \text{female}$ and $credit = \text{bad}$, $check \sim \text{Bern}(0.2)$.
- If checked, $check\text{-failed} = 1$, and from the loan-probability table $P(loan\text{-if-checked} = 1 \mid gender = \text{female}, check\text{-failed} = 1) = 0$.
- If not checked, $loan = 0$ deterministically.

Either branch gives $loan = F$, so

$$\begin{aligned}
PS(gender = \text{female}, loan = F \mid \text{Alice}) \\
&= P(loan_{gender=\text{female}} = F \mid gender = \text{male}, loan = T, credit = \text{bad}) = 1.
\end{aligned}$$

A.3 SHAP value function for Option A

For the 2-feature game $N = \{gender, credit\}$ with $f(g, c) = P(loan = 1 \mid g, c)$, by the SHAP definition (and equivalently by Definition 23 once we note that no dropped feature is a descendant of any coalition member), the value function is

$$v(\mathcal{S}) = \mathbb{E}[f(X) \mid X_{\mathcal{S}} = x_{\mathcal{S}}^*] = \sum_{x_{N \setminus \mathcal{S}} \in \{0,1\}^{|N \setminus \mathcal{S}|}} P(X_{N \setminus \mathcal{S}} = x_{N \setminus \mathcal{S}}) f(x_{\mathcal{S}}^*, x_{N \setminus \mathcal{S}}).$$

Under independent uniform marginals $P(gender) = P(credit) = \frac{1}{2}$ this specialises to

$$\begin{aligned}
v(\emptyset) &= \frac{1}{4} \sum_{g,c \in \{0,1\}} f(g, c), & [\text{both dropped: weight } \frac{1}{2} \cdot \frac{1}{2}] \\
v(\{gender\}) &= \frac{1}{2}[f(g^*, 0) + f(g^*, 1)], & [\text{only credit dropped: weight } P(c) = \frac{1}{2}] \\
v(\{credit\}) &= \frac{1}{2}[f(0, c^*) + f(1, c^*)], & [\text{only gender dropped: weight } P(g) = \frac{1}{2}] \\
v(\{gender, credit\}) &= f(g^*, c^*). & [\text{nothing dropped: no average}]
\end{aligned}$$

Plugging in the four values $f(F, \text{bad}) = 0.000$, $f(F, \text{good}) = 0.180$, $f(M, \text{bad}) = 0.045$, $f(M, \text{good}) = 0.900$ yields the coalition values of Table 4; pushing those through the closed-form Shapley $\phi_i = \frac{1}{2}[v(\{i\}) - v(\emptyset)] + \frac{1}{2}[v(N) - v(N \setminus \{i\})]$ (and flipping signs, since we attribute rejection $g = 1 - f$) gives the magnitudes in Table 5.

B Computations for the Signal-with-Mediation Example

This appendix collects the closed-form derivations for the signal-with-mediation example of Section 4.3: the Gaussian chain $X \rightarrow M \rightarrow Y$ of Eqs. (9)–(11), with $X \sim \mathcal{N}(0.5, 0.25)$, $M = X + \varepsilon_M$, $Y = M + \varepsilon_Y$, and independent noise $\varepsilon_M, \varepsilon_Y \sim \mathcal{N}(0, 0.1)$. The distributional facts and conditional expectations stated in Example 24, and the per-target SHAP and per-pair PCI computations summarised in Table 7, are derived here. Unless noted otherwise the factual instance is Instance 1, $X^* = M^* = Y^* = 1$.

B.1 Distributional facts

From the structural equations, $\text{Var}(M) = \text{Var}(X) + \text{Var}(\varepsilon_M) = 0.25 + 0.1 = 0.35$ and $\text{Var}(Y) = \text{Var}(M) + \text{Var}(\varepsilon_Y) = 0.35 + 0.1 = 0.45$. For the covariances, $\text{Cov}(X, M) = \text{Cov}(X, X + \varepsilon_M) = \text{Var}(X) = 0.25$; $\text{Cov}(X, Y) = \text{Cov}(X, M + \varepsilon_Y) = \text{Cov}(X, M) = 0.25$; and $\text{Cov}(M, Y) = \text{Cov}(M, M + \varepsilon_Y) = \text{Var}(M) = 0.35$.

B.2 Conditional expectations

$\mathbb{E}[Y | X, M] = M$ since $Y = M + \varepsilon_Y$ with $\varepsilon_Y \perp (X, M)$. For $\mathbb{E}[M | X, Y]$, the covariance matrix of (X, Y) is $\Sigma_{XY} = \begin{pmatrix} 0.25 & 0.25 \\ 0.25 & 0.45 \end{pmatrix}$ with $\det(\Sigma_{XY}) = 0.05$, so $\Sigma_{XY}^{-1} = 20 \begin{pmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{pmatrix}$; the regression coefficients are $[\text{Cov}(M, X), \text{Cov}(M, Y)] \Sigma_{XY}^{-1} = [0.25, 0.35] \cdot 20 \begin{pmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{pmatrix} = [0.5, 0.5]$, giving $\mathbb{E}[M | X, Y] = 0.5 + 0.5(X - 0.5) + 0.5(Y - 0.5) = 0.5X + 0.5Y$. For $\mathbb{E}[X | M, Y]$, by d -separation, $X \perp Y | M$ in the chain $X \rightarrow M \rightarrow Y$, so $\mathbb{E}[X | M, Y] = \mathbb{E}[X | M] = 0.5 + \frac{\text{Cov}(X, M)}{\text{Var}(M)}(M - 0.5) = 0.5 + \frac{0.25}{0.35}(M - 0.5) = 0.5 + \frac{5}{7}(M - 0.5)$.

B.3 Plain SHAP at Instance 1

Target Y , features $\{X, M\}$. First, compute the coalition values:

$$v(\emptyset) = \mathbb{E}[M] = 0.5 \quad (22)$$

$$v(\{X\}) = \mathbb{E}[M | X = 1] = 1 \quad (23)$$

$$v(\{M\}) = \mathbb{E}[M | M = 1] = 1 \quad (24)$$

$$v(\{X, M\}) = f_Y(1, 1) = 1 \quad (25)$$

Next, apply equation (5) with $|N| = 2$:

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 1) = 0.25 \quad \phi_M = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 1) = 0.25 \quad (26)$$

Target M , features $\{X, Y\}$. $v(\emptyset) = 0.5$; $v(\{X\}) = 1$; $v(\{Y\}) = 0.5 \mathbb{E}[X | Y=1] + 0.5 = 0.889$ (using $\mathbb{E}[X | Y=1] = 0.5 + \frac{0.25}{0.45}(0.5) = 0.778$); $v(\{X, Y\}) = 1$.

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 0.889) = 0.306, \quad (27)$$

$$\phi_Y = \frac{1}{2}(0.889 - 0.5) + \frac{1}{2}(1 - 1) = 0.194. \quad (28)$$

Target X , features $\{M, Y\}$. $v(\emptyset) = 0.5$; $v(\{M\}) = 0.5 + \frac{5}{7}(0.5) = 0.857$; $v(\{Y\}) = \mathbb{E}[X | Y=1] = 0.778$ (as above); $v(\{M, Y\}) = \mathbb{E}[X | M=1] = 0.857$ (since $X \perp Y | M$).

$$\phi_M = \frac{1}{2}(0.857 - 0.5) + \frac{1}{2}(0.857 - 0.778) = 0.219, \quad (29)$$

$$\phi_Y = \frac{1}{2}(0.778 - 0.5) + \frac{1}{2}(0.857 - 0.857) = 0.139. \quad (30)$$

B.4 Causal SHAP at Instance 1

We apply Causal SHAP (Definition 23) and compare against plain SHAP.

Target Y , features $\{X, M\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = \mathbb{E}[M | \text{do}(X=1)] = 1$ (M is downstream of X : $P(M | \text{do}(X=1)) = \mathcal{N}(1, 0.1)$); $v_{\text{causal}}(\{M\}) = 1$ ($\text{do}(M=1)$ sets $f_Y = M = 1$ directly, $P(X | \text{do}(M=1)) = P(X)$ since X has no causal parents); $v_{\text{causal}}(\{X, M\}) = 1$. All values match plain SHAP, so $\phi_X = \phi_M = 0.25$.

Target M , features $\{X, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = 1$ (Y is downstream of X , same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{X \sim P(X)}[0.5X + 0.5 \cdot 1] = 0.75$ (Y does not cause X , so $P(X | \text{do}(Y=1)) = P(X)$); differs from plain SHAP's 0.889; $v_{\text{causal}}(\{X, Y\}) = 1$.

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 0.75) = 0.375, \quad (31)$$

$$\phi_Y = \frac{1}{2}(0.75 - 0.5) + \frac{1}{2}(1 - 1) = 0.125. \quad (32)$$

Target X , features $\{M, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{M\}) = 0.857$ (Y is downstream of M but f_X does not depend on Y since $X \perp Y | M$, so $\mathbb{E}_{Y \sim P(Y | \text{do}(M=1))}[f_X(1, Y)] = f_X(1, \cdot) = 0.857$;

same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{M \sim P(M)}[f_X(M, 1)] = 0.5$ (M is not a descendant of Y , so $P(M \mid \text{do}(Y=1)) = P(M)$ and $\mathbb{E}[f_X(M, 1)] = 0.5 + \frac{5}{7}(\mathbb{E}[M] - 0.5) = 0.5$; differs from plain SHAP's 0.778); $v_{\text{causal}}(\{M, Y\}) = 0.857$.

$$\phi_M = \frac{1}{2}(0.857 - 0.5) + \frac{1}{2}(0.857 - 0.5) = 0.357, \quad (33)$$

$$\phi_Y = \frac{1}{2}(0.5 - 0.5) + \frac{1}{2}(0.857 - 0.857) = 0. \quad (34)$$

B.5 PCI at Instance 1

We work at the factual instance $X^* = M^* = Y^* = 1$ with the absolute-difference score (Example 21): $ci(y^s, y^n, y^*) = |y^n - y^*| - |y^s - y^*|$. Both worlds run the PSCM with fresh exogenous noise drawn from P_U : the necessity world replaces the suspect A with $A' \sim P(A)$ and propagates; the sufficiency world restores it to a^* and propagates. For target B , $\mathbf{S}$ is the two input features of B and $\mathbf{W}$ the remaining input; Γ_s is uniform over the three non-empty subsets of $\mathbf{S}$ and Γ_w uniform over $2^{\mathbf{W}}$.

$R(X \rightsquigarrow Y)$. $\mathbf{S} = \{X, M\}$, $\mathbf{W} = \{M\}$, $w^* = m^* = 1$, and $X' \sim \mathcal{N}(0.5, 0.25)$. Three of the four accepted $(\mathbf{C}, \mathbf{T})$ pairs intervene on X , each with weight $\frac{1}{4}$; the fourth, $(\mathbf{C}=\{M\}, \mathbf{T}=\emptyset)$, does not touch X and contributes nothing. The key observation is that the witness pair $(\mathbf{C}=\{X\}, \mathbf{T}=\{M\})$ pins M at m^* in both worlds, which equalizes their distributions and zeros out $\bar{c}$; the other two pairs let M propagate freely. Computing $\mathbb{E}[|Y^n - 1|]$ and $\mathbb{E}[|Y^s - 1|]$ in closed form via the folded-normal formula gives:

$(\mathbf{C}, \mathbf{T})$	$\mathbb{E}[Y^n - 1]$	$\mathbb{E}[Y^s - 1]$	$\bar{c}$
$(\{X\}, \emptyset)$	0.677	0.357	0.320
$(\{X\}, \{M\})$ [witness]	0.252	0.252	0.000
$(\{X, M\}, \emptyset)$	0.677	0.252	0.425

Combining at weight $\frac{1}{4}$ each,

$$R(X \rightsquigarrow Y \mid \mathbf{W}=\{M\}) = \frac{1}{4}(0.320 + 0 + 0.425) \approx 0.186.$$

$R(Y \rightsquigarrow X)$. X is exogenous: no *do*-intervention on $\{M, Y\}$ alters its structural equation, so necessity and sufficiency distributions coincide and $R(Y \rightsquigarrow X \mid \mathbf{W}=\{M\}) = 0$.

Other cells. The remaining entries of Table 7 (D-MY, D-XM, D-YM, D-MX, D-MXY, and the $\mathbf{W} = \emptyset$ column) follow the same closed-form machinery; full per-pair computations are in the companion notebook `signal_mediation_computations.ipynb`.

C Setup for the Desert-Traveller PCI Computation

This appendix records the structural model and the full PCI specification behind the desert-traveller comparison of Section 8. The basic-variant results (Table 12) and the weak-poison results (Table 14) are produced by running the PCI Monte Carlo search of Section 3 on the scenario-specific model below, with the component choices listed here; a parallel hand-rolled enumeration matches every value within Monte Carlo noise (companion notebook `docs/source/desert_traveler.ipynb`).

C.1 Structural model (basic variant)

To compare directly with Pearl, we run PCI under the same forensic conditioning: one structural model per scenario, with u pinned at the scenario's forensic realisation. The endogenous variables are $X, P, c, d, Y \in \{0, 1\}$ with

$$c = P(u' \vee X'), \quad d = X(u \vee P'), \quad y = c \vee d,$$

where primes denote Boolean complement ($u' = 1 - u$, etc.), c indicates the cyanide path operating (poisoned water drunk) and d the dehydration path operating (drank from empty canteen).

C.2 PCI components

The PCI components, all specified in advance:

- Suspects $\mathbf{S} = \{X, P\}$; candidate causes are the singletons $\{X\}$ and $\{P\}$.
- Witness pool $\mathbf{W} = \{c, d\}$ (the mediators); four subsets $\mathbf{T} \subseteq \mathbf{W}$, weighted by Γ cardinality-uniform (uniform on $|\mathbf{T}|$, then uniform within each cardinality class).
- Non-cause suspect (the one not in $\mathbf{C}$) is *not* intervened on or hard-coded; it follows the SCM under the noise prior on its own coin, consistent with the formal definition’s requirement that only $\mathbf{C}$ and $\mathbf{T}$ are intervened (Definition 18).
- Alternative-value distribution Δ deterministically flips $1 \rightarrow 0$.
- Noise distribution P_U is the forensic posterior, a point mass at the scenario’s u . We do *not* condition on the outcome Y itself; the forensic evidence sits above Y in the SCM, so conditioning on it is not the “fully conditional” anti-pattern of Section 3.

For the weak-poison variant (Section 8.4) the only mechanical changes are that the witness pool becomes $\mathbf{W} = \{c, d, v_C\}$ (the new “cyanide-was-fatal” mediator v_C is included) and that the Scenario A forensic posterior pins both $u = 0$ and $\xi = 1$; Scenario B’s posterior pins $u = 1$, with ξ structurally irrelevant since the cyanide path is dormant.